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Lee et al.

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(54) **METHOD AND APPARATUS FOR TRANSMITTING BUFFER STATUS REPORT BY WIRELESS NODE IN WIRELESS COMMUNICATION SYSTEM**

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(60) Provisional application No. 62/775,789, filed on Dec. 5, 2018.

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H04W 72/1263 (2023.01)
H04W 72/566 (2023.01)
H04W 80/02 (2009.01)

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CPC **H04W 72/21** (2023.01); **H04W 72/1263** (2013.01); **H04W 72/569** (2023.01); **H04W 80/02** (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

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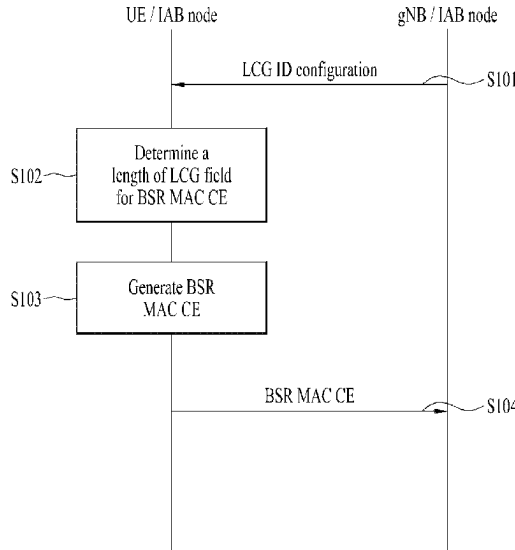
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(57) **ABSTRACT**

The present invention relates to a method of transmitting a buffer status report (BSR) by a wireless node in a wireless communication system. In particular, the method includes the steps of: receiving first logical channel group (LCG) configuration information including identities of LCGs from a network; generating the BSR including a LCG field; and transmitting the BSR to the network, wherein a length of the LCG field is configured according to a highest value of the identities of LCGs.

10 Claims, 11 Drawing Sheets



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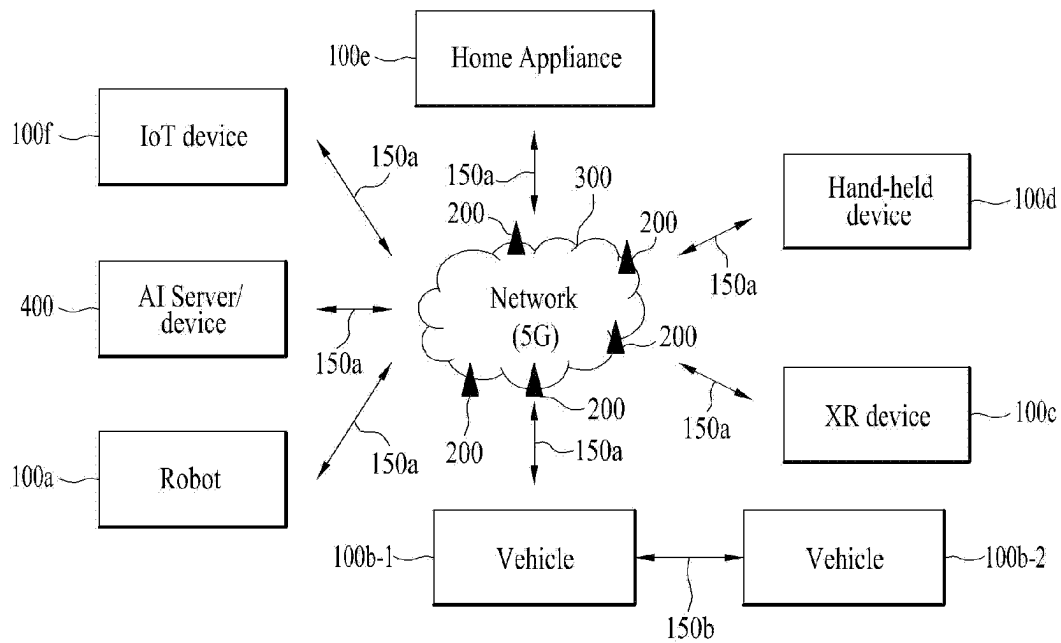
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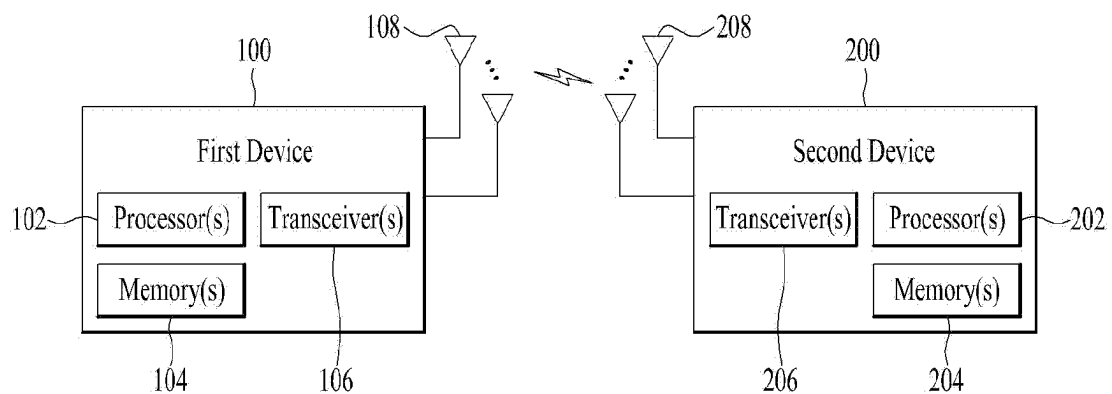
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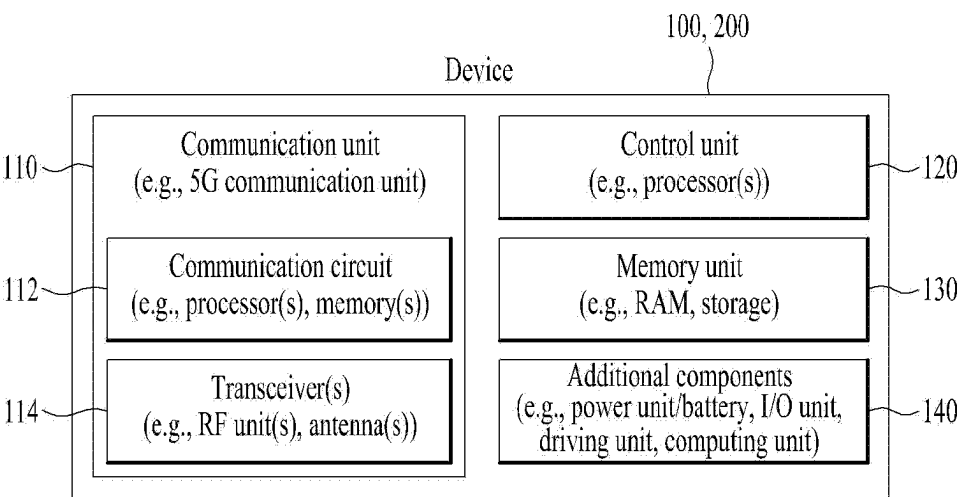
[Fig. 1]



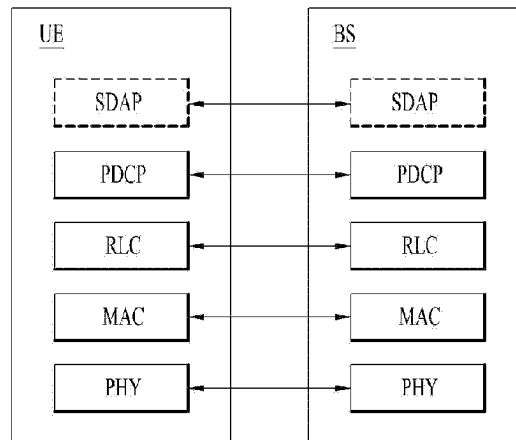
[Fig. 2]



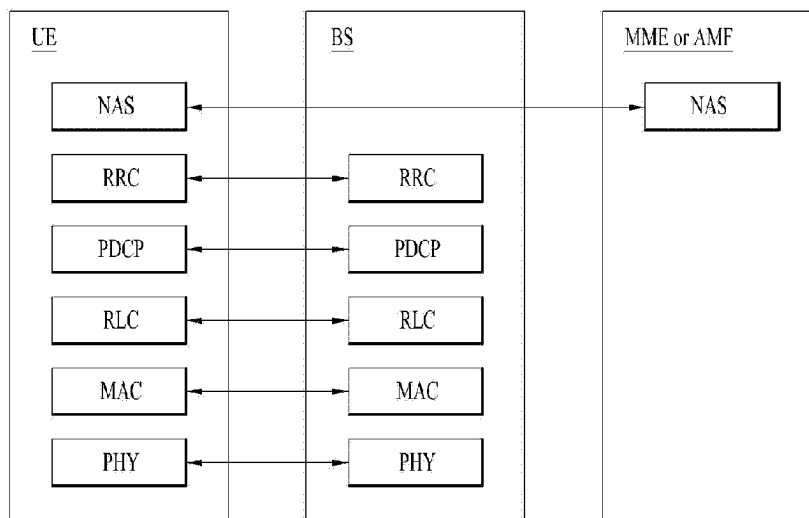
[Fig. 3]



[Fig. 4]

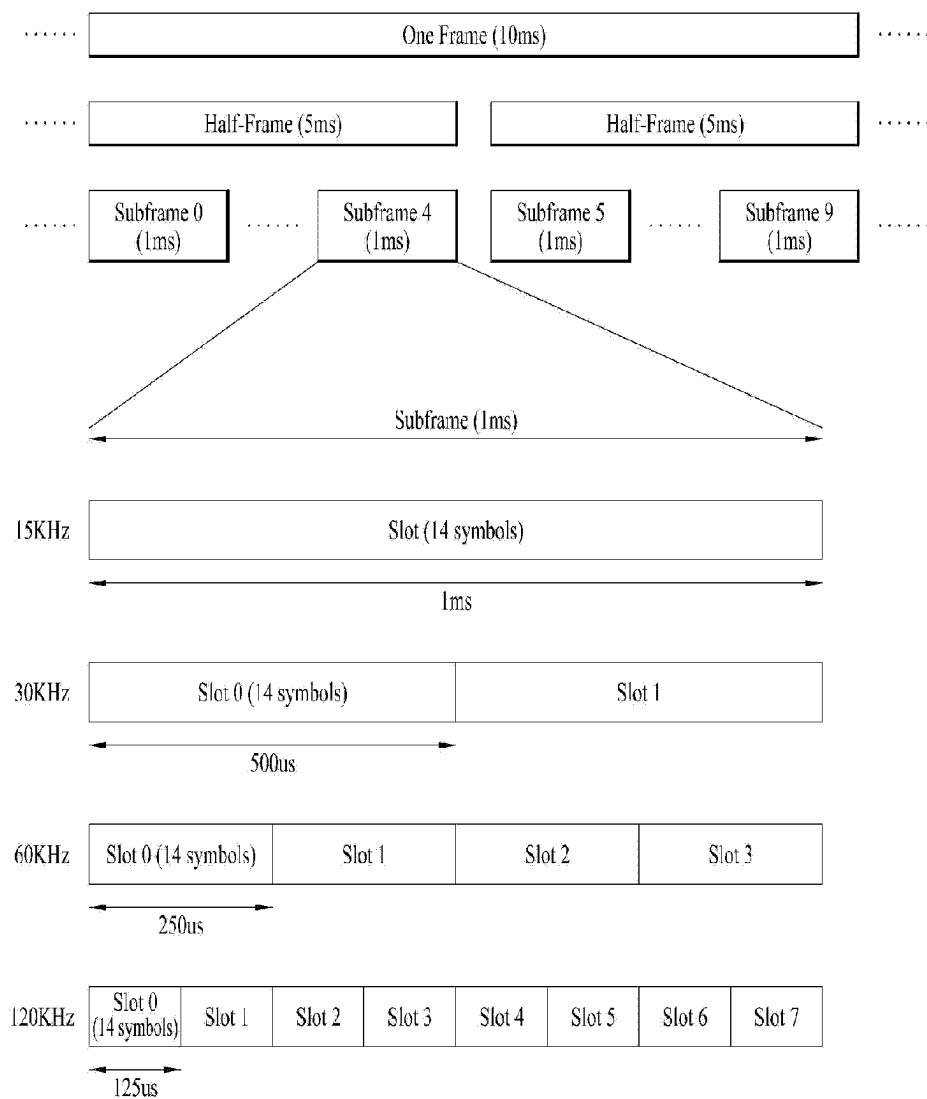


(a) User Plane Protocol Stack

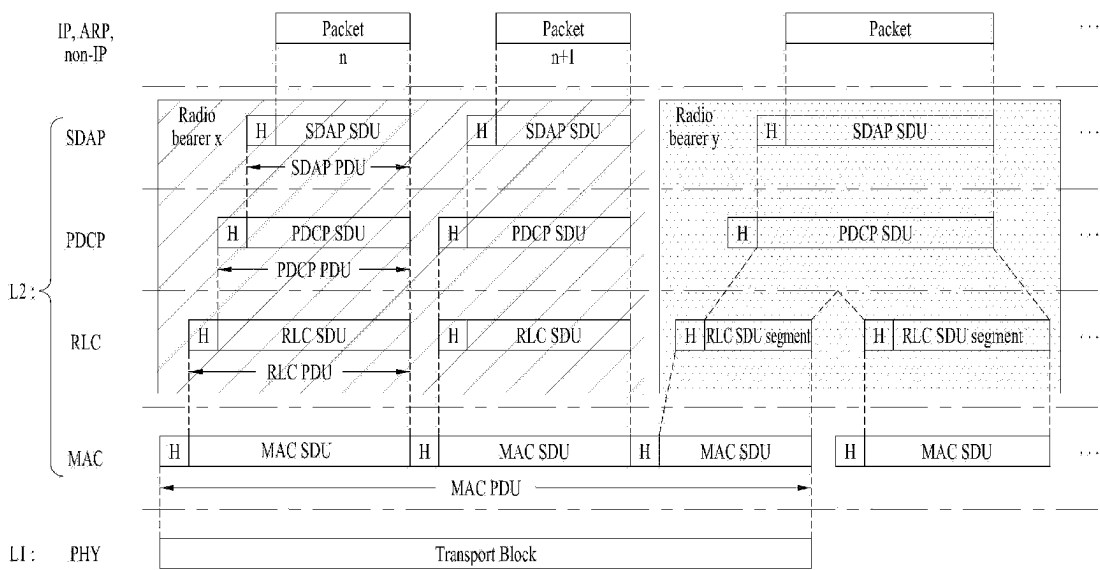


(b) Control Plane Protocol Stack

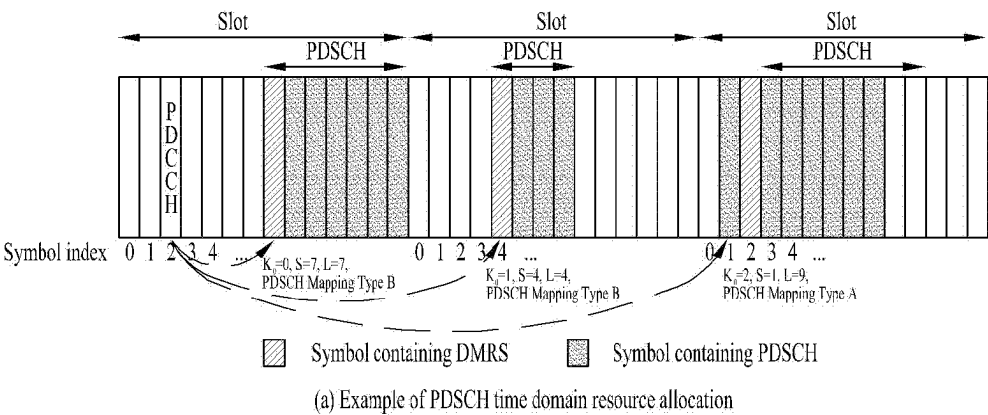
[Fig. 5]



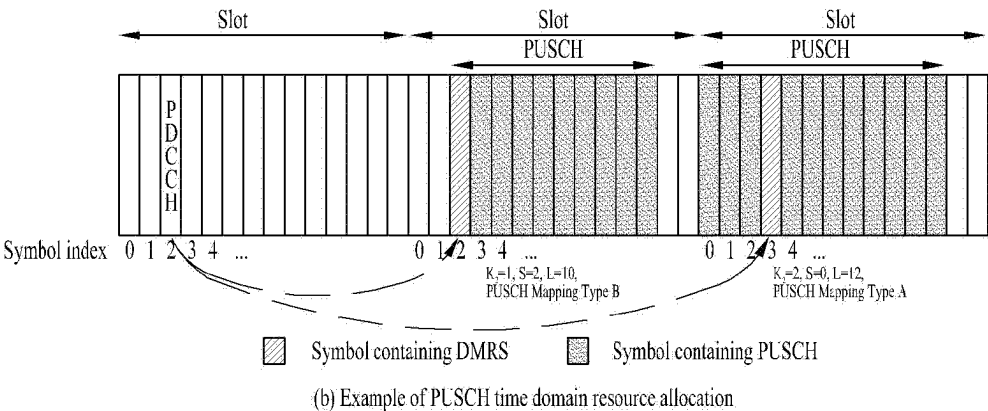
[Fig. 6]



[Fig. 7]

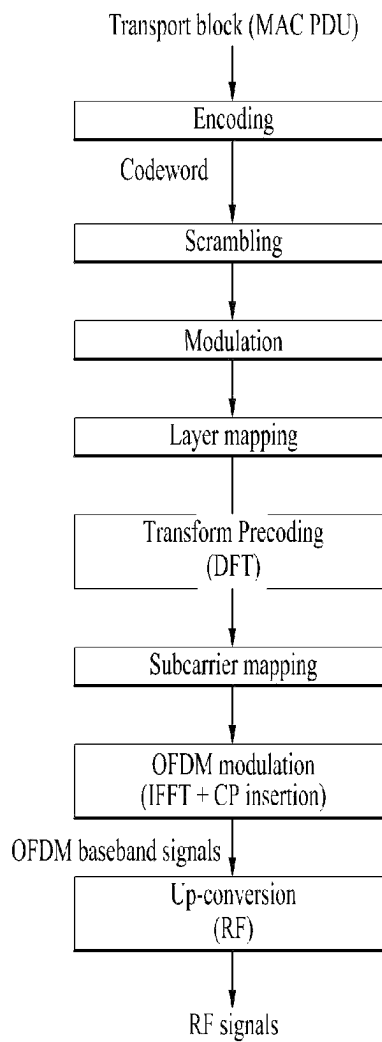


(a) Example of PDSCH time domain resource allocation

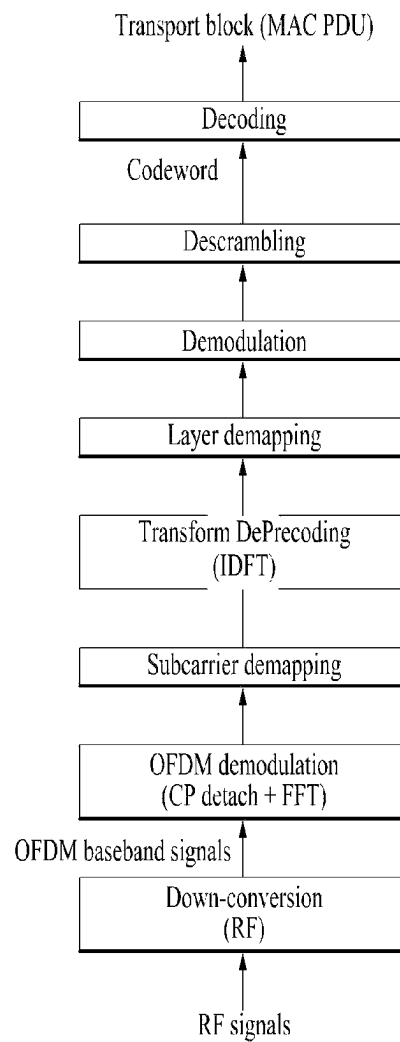


(b) Example of PUSCH time domain resource allocation

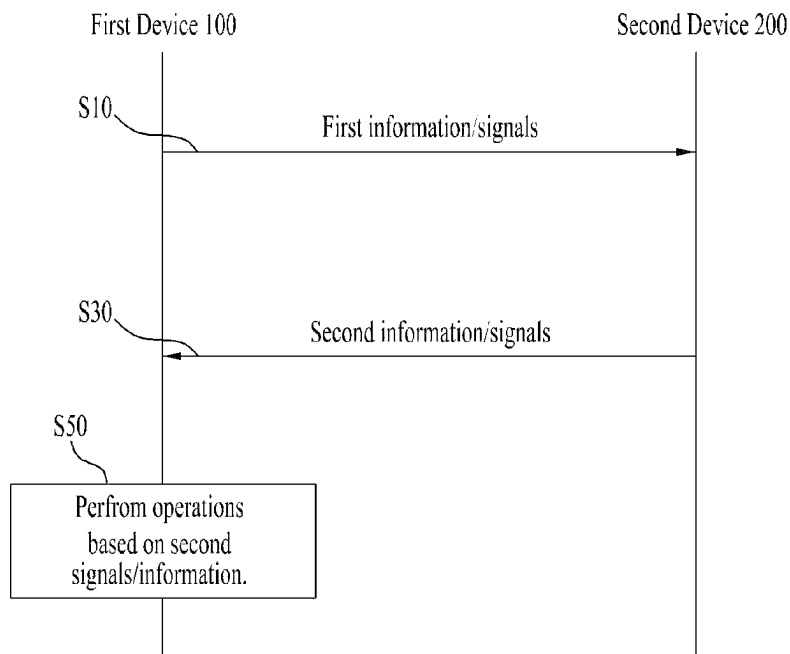
[Fig. 8]



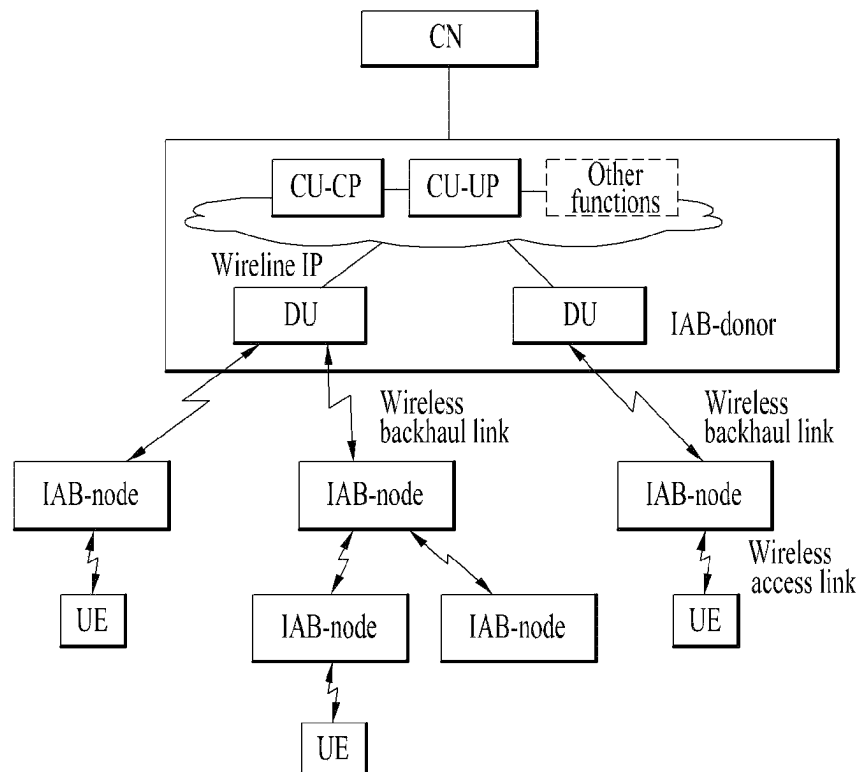
[Fig. 9]



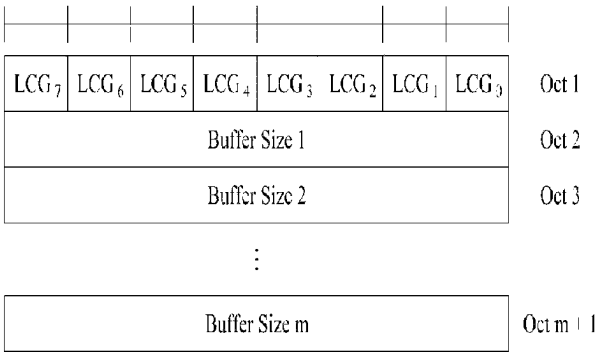
[Fig. 10]



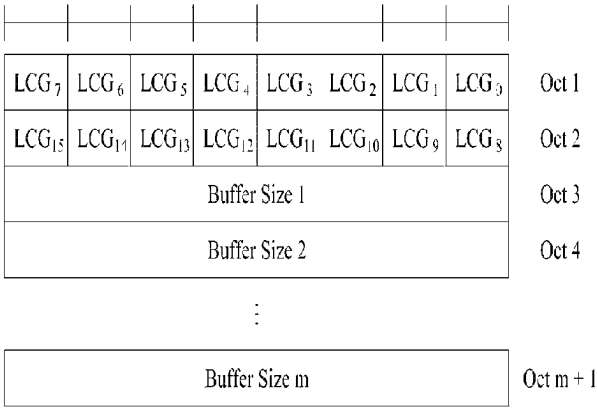
[Fig. 11]



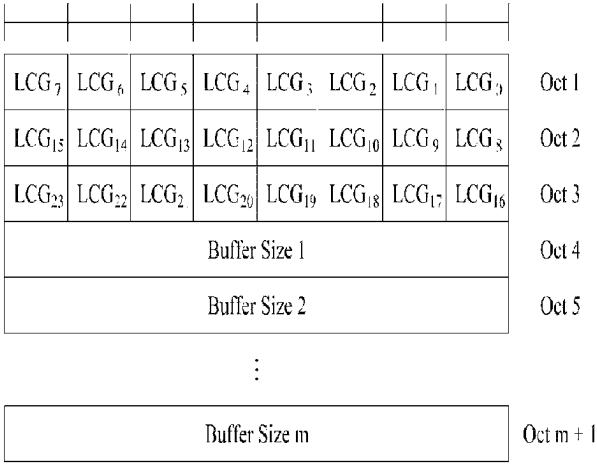
[Fig. 12]



(a)

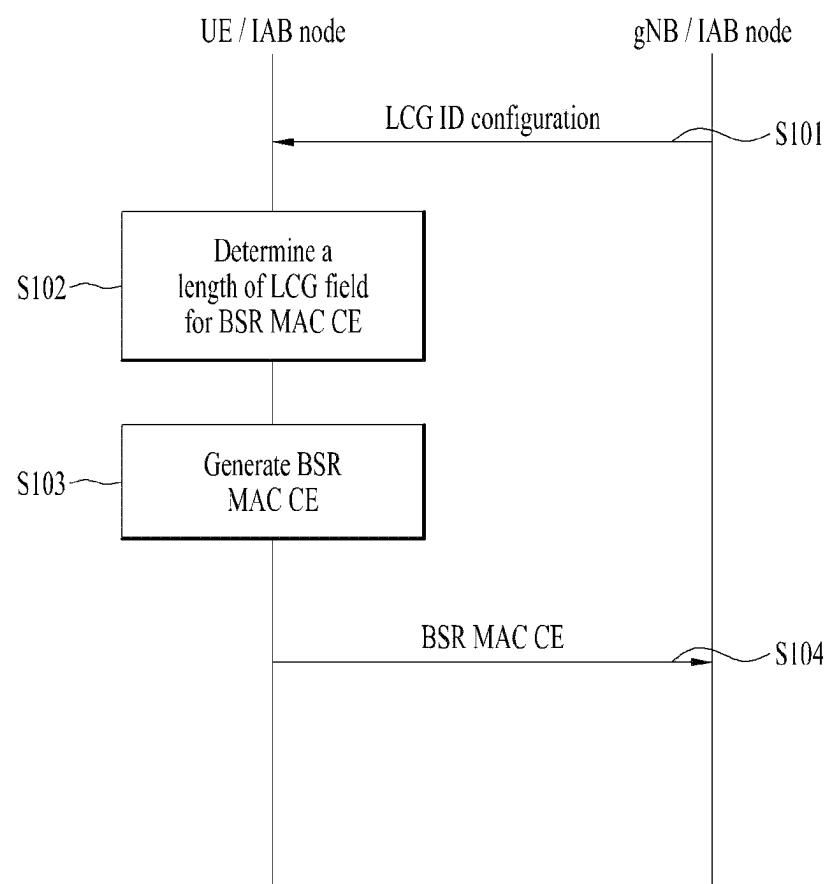


(b)

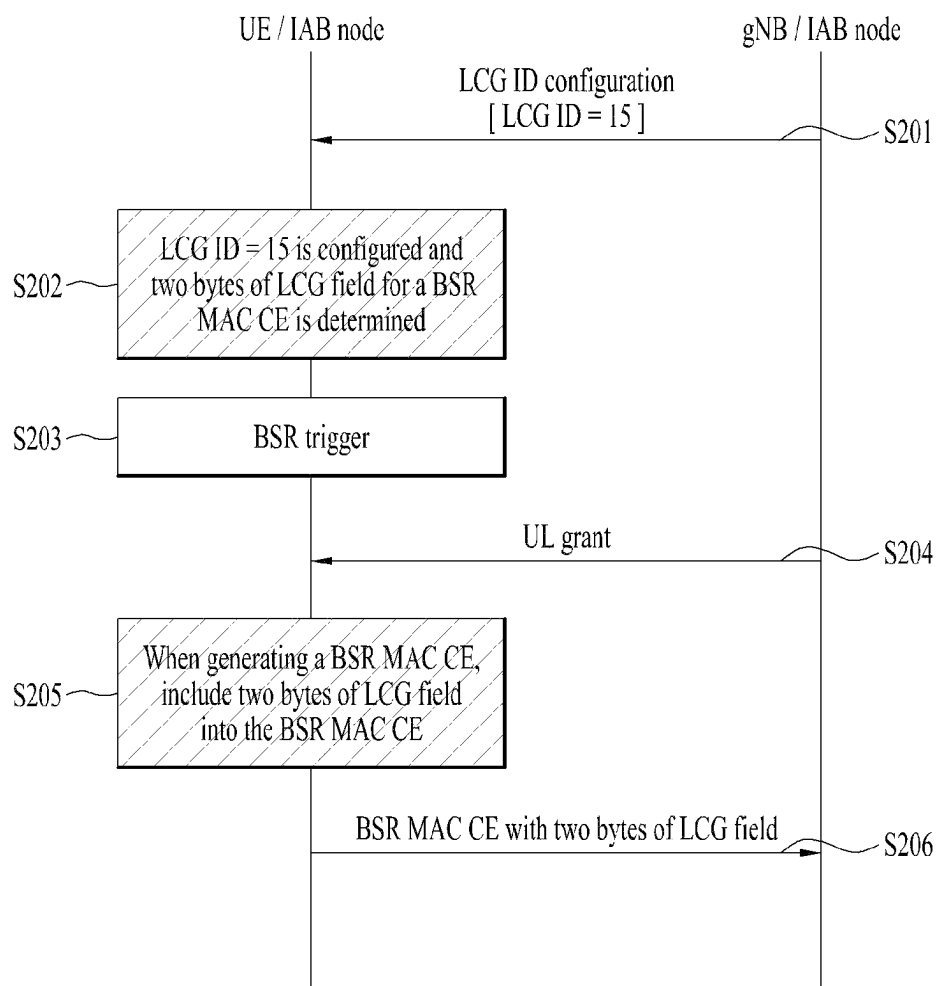


(c)

[Fig. 13]



[Fig. 14]



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METHOD AND APPARATUS FOR TRANSMITTING BUFFER STATUS REPORT BY WIRELESS NODE IN WIRELESS COMMUNICATION SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/283,907, filed on Apr. 8, 2021, which is the National Stage filing under 35 U.S.C. 371 of International Application No. PCT/KR2019/015796, filed on Nov. 19, 2019, which claims the benefit of U.S. Provisional Application No. 62/775,789, filed on Dec. 5, 2018, the contents of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

The present invention relates to a wireless communication system, and more particularly, to a method for transmitting a buffer status report (BSR) in a wireless communication system and an apparatus therefor.

BACKGROUND ART

Introduction of new radio communication technologies has led to increases in the number of user equipments (UEs) to which a base station (BS) provides services in a prescribed resource region, and has also led to increases in the amount of data and control information that the BS transmits to the UEs. Due to typically limited resources available to the BS for communication with the UE(s), new techniques are needed by which the BS utilizes the limited radio resources to efficiently receive/transmit uplink/downlink data and/or uplink/downlink control information. In particular, overcoming delay or latency has become an important challenge in applications whose performance critically depends on delay/latency

DISCLOSURE OF INVENTION

Technical Problem

Accordingly, an object of the present invention is to provide a method performing measurement by a user equipment (UE) in a wireless communication system and an apparatus therefor.

Solution to Problem

The object of the present invention can be achieved by the method for transmitting a buffer status report (BSR) by a wireless node in a wireless communication system comprising steps of receiving first logical channel group (LCG) configuration information including identities of LCGs from a network; generating the BSR including a LCG field; and transmitting the BSR to the network, wherein a length of the LCG field is configured according to a highest value of the identities of LCGs.

Further, it is suggested a wireless node in a wireless communication system comprising a memory; and at least one processor coupled to the memory. More specifically, the at least one processor is configured to, receive first logical channel group (LCG) configuration information including identities of LCGs from a network; generate a buffer status report (BSR) including a LCG field; and transmit the BSR

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to the network, wherein a length of the LCG field is configured according to a highest value of the identities of LCGs.

Preferably, when second LCG configuration information is received from the network, the length of the LCG field is reconfigured according to a highest value of the identities of LCGs included in the second LCG configuration information.

Preferably, the length of the LCG field is configured to floor(Y/8) plus 1 bytes, when highest value of the identities of LCGs is equal to or larger than Y and less than Y+8 (where Y is multiple of 8).

Preferably, each bit of the LCG field indicates whether buffer size information of a corresponding LCG is present or not in the BSR.

Preferably, a number of the identities of LCGs is also variable according to a number of node or flow/radio bearer served by the wireless node.

Preferably, the at least one processor is further configured to implement at least one advanced driver assistance system (ADAS) function based on signals that control the wireless node.

Advantageous Effects of Invention

According to the aforementioned embodiments of the present invention, the wireless node can transmit the BSR efficiently.

Effects obtainable from the present invention may be non-limited by the above mentioned effect. And, other unmentioned effects can be clearly understood from the following description by those having ordinary skill in the technical field to which the present invention pertains.

BRIEF DESCRIPTION OF DRAWINGS

The accompanying drawings, which are included to provide a further understanding of the invention, illustrate embodiments of the invention and together with the description serve to explain the principle of the invention:

FIG. 1 illustrates an example of a communication system 1 to which implementations of the present disclosure is applied;

FIG. 2 is a block diagram illustrating examples of communication devices which can perform a method according to the present disclosure;

FIG. 3 illustrates another example of a wireless device which can perform implementations of the present invention;

FIG. 4 illustrates an example of protocol stacks in a third generation partnership project (3GPP) based wireless communication system;

FIG. 5 illustrates an example of a frame structure in a 3GPP based wireless communication system;

FIG. 6 illustrates a data flow example in the 3GPP new radio (NR) system;

FIG. 7 illustrates an example of PDSCH time domain resource allocation by PDCCH, and an example of PUSCH time resource allocation by PDCCH;

FIG. 8 illustrates an example of physical layer processing at a transmitting side;

FIG. 9 illustrates an example of physical layer processing at a receiving side.

FIG. 10 illustrates operations of the wireless devices based on the implementations of the present disclosure;

FIG. 11 shows an example of IAB based RAN architectures;

FIG. 12 shows examples of eBSR format according to the present disclosure;

FIG. 13 shows an example of a BSR procedure according to the present disclosure; and

FIG. 14 shows another example of a BSR procedure according to the present disclosure.

BEST MODE FOR CARRYING OUT THE INVENTION

Reference will now be made in detail to the exemplary implementations of the present disclosure, examples of which are illustrated in the accompanying drawings. The detailed description, which will be given below with reference to the accompanying drawings, is intended to explain exemplary implementations of the present disclosure, rather than to show the only implementations that can be implemented according to the disclosure. The following detailed description includes specific details in order to provide a thorough understanding of the present disclosure. However, it will be apparent to those skilled in the art that the present disclosure may be practiced without such specific details.

The following techniques, apparatuses, and systems may be applied to a variety of wireless multiple access systems. Examples of the multiple access systems include a code division multiple access (CDMA) system, a frequency division multiple access (FDMA) system, a time division multiple access (TDMA) system, an orthogonal frequency division multiple access (OFDMA) system, a single carrier frequency division multiple access (SC-FDMA) system, and a multicarrier frequency division multiple access (M C-FDMA) system. CDMA may be embodied through radio technology such as universal terrestrial radio access (UTRA) or CDMA2000. TDMA may be embodied through radio technology such as global system for mobile communications (GSM), general packet radio service (GPRS), or enhanced data rates for GSM evolution (EDGE). OFDMA may be embodied through radio technology such as institute of electrical and electronics engineers (IEEE) 802.11 (Wi-Fi), IEEE 802.16 (WiMAX), IEEE 802.20, or evolved UTRA (E-UTRA). UTRA is a part of a universal mobile telecommunications system (UMTS). 3rd generation partnership project (3GPP) long term evolution (LTE) is a part of evolved UMTS (E-UMTS) using E-UTRA. 3GPP LTE employs OFDMA in DL and SC-FDMA in UL. LTE-advanced (LTE-A) is an evolved version of 3GPP LTE.

For convenience of description, implementations of the present disclosure are mainly described in regards to a 3GPP based wireless communication system. However, the technical features of the present disclosure are not limited thereto. For example, although the following detailed description is given based on a mobile communication system corresponding to a 3GPP based wireless communication system, aspects of the present disclosure that are not limited to 3GPP based wireless communication system are applicable to other mobile communication systems. For terms and technologies which are not specifically described among the terms of and technologies employed in the present disclosure, the wireless communication standard documents published before the present disclosure may be referenced. For example, the following documents may be referenced.

3GPP LTE

3GPP TS 36.211: Physical channels and modulation
3GPP TS 36.212: Multiplexing and channel coding
3GPP TS 36.213: Physical layer procedures
3GPP TS 36.214: Physical layer; Measurements

3GPP TS 36.300: Overall description

3GPP TS 36.304: User Equipment (UE) procedures in idle mode

3GPP TS 36.314: Layer 2—Measurements

3GPP TS 36.321: Medium Access Control (MAC) protocol

3GPP TS 36.322: Radio Link Control (RLC) protocol

3GPP TS 36.323: Packet Data Convergence Protocol (PDCP)

3GPP TS 36.331: Radio Resource Control (RRC) protocol

3GPP NR (e.g. 5G)

3GPP TS 38.211: Physical channels and modulation

3GPP TS 38.212: Multiplexing and channel coding

3GPP TS 38.213: Physical layer procedures for control

3GPP TS 38.214: Physical layer procedures for data

3GPP TS 38.215: Physical layer measurements

3GPP TS 38.300: Overall description

3GPP TS 38.304: User Equipment (UE) procedures in idle mode and in RRC inactive state

3GPP TS 38.321: Medium Access Control (MAC) protocol

3GPP TS 38.322: Radio Link Control (RLC) protocol

3GPP TS 38.323: Packet Data Convergence Protocol (PDCP)

3GPP TS 38.331: Radio Resource Control (RRC) protocol

3GPP TS 37.324: Service Data Adaptation Protocol (SDAP)

3GPP TS 37.340: Multi-connectivity; Overall description

In the present disclosure, a user equipment (UE) may be a fixed or mobile device. Examples of the UE include various devices that transmit and receive user data and/or various kinds of control information to and from a base station (BS). In the present disclosure, a BS generally refers to a fixed station that performs communication with a UE and/or another BS, and exchanges various kinds of data and control information with the UE and another BS. The BS may be referred to as an advanced base station (ABS), a node-B (NB), an evolved node-B (eNB), a base transceiver system (BTS), an access point (AP), a processing server (PS), etc. Especially, a BS of the UMTS is referred to as a NB, a BS of the enhanced packet core (EPC)/long term evolution (LTE) system is referred to as an eNB, and a BS of the new radio (NR) system is referred to as a gNB.

In the present disclosure, a node refers to a point capable of transmitting/receiving a radio signal through communication with a UE. Various types of BSs may be used as nodes irrespective of the terms thereof. For example, a BS, a node B (NB), an e-node B (eNB), a pico-cell eNB (PeNB), a home eNB (HeNB), a relay, a repeater, etc. may be a node. In addition, the node may not be a BS. For example, the node may be a radio remote head (RRH) or a radio remote unit (RRU). The RRH or RRU generally has a lower power level than a power level of a BS. Since the RRH or RRU (hereinafter, RRH/RRU) is generally connected to the BS through a dedicated line such as an optical cable, cooperative communication between RRH/RRU and the BS can be smoothly performed in comparison with cooperative communication between BSs connected by a radio line. At least one antenna is installed per node. The antenna may include a physical antenna or an antenna port or a virtual antenna.

In the present disclosure, the term “cell” may refer to a geographic area to which one or more nodes provide a communication system, or refer to radio resources. A “cell” of a geographic area may be understood as coverage within which a node can provide service using a carrier and a “cell”

as radio resources (e.g. time-frequency resources) is associated with bandwidth (BW) which is a frequency range configured by the carrier. The “cell” associated with the radio resources is defined by a combination of downlink resources and uplink resources, for example, a combination of a downlink (DL) component carrier (CC) and an uplink (UL) CC. The cell may be configured by downlink resources only, or may be configured by downlink resources and uplink resources. Since DL coverage, which is a range within which the node is capable of transmitting a valid signal, and UL coverage, which is a range within which the node is capable of receiving the valid signal from the UE, depends upon a carrier carrying the signal, the coverage of the node may be associated with coverage of the “cell” of radio resources used by the node. Accordingly, the term “cell” may be used to represent service coverage of the node sometimes, radio resources at other times, or a range that signals using the radio resources can reach with valid strength at other times.

In the present disclosure, a physical downlink control channel (PDCCH), and a physical downlink shared channel (PDSCH) refer to a set of time-frequency resources or resource elements (REs) carrying downlink control information (DCI), and a set of time-frequency resources or REs carrying downlink data, respectively. In addition, a physical uplink control channel (PUCCH), a physical uplink shared channel (PUSCH) and a physical random access channel (PRACH) refer to a set of time-frequency resources or REs carrying uplink control information (UCI), a set of time-frequency resources or REs carrying uplink data and a set of time-frequency resources or REs carrying random access signals, respectively.

In carrier aggregation (CA), two or more CCs are aggregated. A UE may simultaneously receive or transmit on one or multiple CCs depending on its capabilities. CA is supported for both contiguous and non-contiguous CCs. When CA is configured the UE only has one radio resource control (RRC) connection with the network. At RRC connection establishment/re-establishment/handover, one serving cell provides the non-access stratum (NAS) mobility information, and at RRC connection re-establishment/handover, one serving cell provides the security input. This cell is referred to as the Primary Cell (PCell). The PCell is a cell, operating on the primary frequency, in which the UE either performs the initial connection establishment procedure or initiates the connection re-establishment procedure. Depending on UE capabilities, Secondary Cells (SCells) can be configured to form together with the PCell a set of serving cells. An SCell is a cell providing additional radio resources on top of Special Cell. The configured set of serving cells for a UE therefore always consists of one PCell and one or more SCells. In the present disclosure, for dual connectivity (DC) operation, the term “special Cell” refers to the PCell of the master cell group (MCG) or the PCell of the secondary cell group (SCG), and otherwise the term Special Cell refers to the PCell. An SpCell supports physical uplink control channel (PUCCH) transmission and contention-based random access, and is always activated. The MCG is a group of serving cells associated with a master node, comprising of the SpCell (PCell) and optionally one or more SCells. The SCG is the subset of serving cells associated with a secondary node, comprising of the PCell and zero or more SCells, for a UE configured with DC. For a UE in RRC_CONNECTED not configured with CA/DC there is only one serving cell comprising of the PCell. For a UE in RRC_CO-

NNECTED configured with CA/DC the term “serving cells” is used to denote the set of cells comprising of the SpCell(s) and all SCells.

The MCG is a group of serving cells associated with a master BS which terminates at least S1-MME, and the SCG is a group of serving cells associated with a secondary BS that is providing additional radio resources for the UE but is not the master BS. The SCG includes a primary SCell (PSCell) and optionally one or more SCells. In DC, two MAC entities are configured in the UE: one for the MCG and one for the SCG. Each MAC entity is configured by RRC with a serving cell supporting PUC CH transmission and contention based Random Access. In the present disclosure, the term SpCell refers to such cell, whereas the term SCell refers to other serving cells. The term SpCell either refers to the PCell of the MCG or the PSCell of the SCG depending on if the MAC entity is associated to the MCG or the SCG, respectively.

In the present disclosure, monitoring a channel refers to attempting to decode the channel. For example, monitoring a physical downlink control channel (PDCCH) refers to attempting to decode PDCCH(s) (or PDCCH candidates).

In the present disclosure, “C-RNTI” refers to a cell RNTI, “SI-RNTI” refers to a system information RNTI, “P-RNTI” refers to a paging RNTI, “RA-RNTI” refers to a random access RNTI, “SC-RNTI” refers to a single cell RNTI, “SL-RNTI” refers to a sidelink RNTI, “SPS C-RNTI” refers to a semi-persistent scheduling C-RNTI, and “CS-RNTI” refers to a configured scheduling RNTI.

FIG. 1 illustrates an example of a communication system 1 to which implementations of the present disclosure is applied.

Three main requirement categories for 5G include (1) a category of enhanced mobile broadband (eMBB), (2) a category of massive machine type communication (mMTC), and (3) a category of ultra-reliable and low latency communications (URLLC).

Partial use cases may require a plurality of categories for optimization and other use cases may focus only upon one key performance indicator (KPI). 5G supports such various use cases using a flexible and reliable method.

eMBB far surpasses basic mobile Internet access and covers abundant bidirectional work and media and entertainment applications in cloud and augmented reality. Data is one of 5G core motive forces and, in a 5G era, a dedicated voice service may not be provided for the first time. In 5G, it is expected that voice will be simply processed as an application program using data connection provided by a communication system. Main causes for increased traffic volume are due to an increase in the size of content and an increase in the number of applications requiring high data transmission rate. A streaming service (of audio and video), conversational video, and mobile Internet access will be more widely used as more devices are connected to the Internet. These many application programs require connectivity of an always turned-on state in order to push real-time information and alarm for users. Cloud storage and applications are rapidly increasing in a mobile communication platform and may be applied to both work and entertainment. The cloud storage is a special use case which accelerates growth of uplink data transmission rate. 5G is also used for remote work of cloud. When a tactile interface is used, 5G demands much lower end-to-end latency to maintain user good experience. Entertainment, for example, cloud gaming and video streaming, is another core element which increases demand for mobile broadband capability. Entertainment is essential for a smartphone and a tablet in

anyplace including high mobility environments such as a train, a vehicle, and an airplane. Other use cases are augmented reality for entertainment and information search. In this case, the augmented reality requires very low latency and instantaneous data volume.

In addition, one of the most expected 5G use cases relates a function capable of smoothly connecting embedded sensors in all fields, i.e., mMTC. It is expected that the number of potential IoT devices will reach 204 hundred million up to the year of 2020. An industrial IoT is one of categories of performing a main role enabling a smart city, asset tracking, smart utility, agriculture, and security infrastructure through 5G.

URLLC includes a new service that will change industry through remote control of main infrastructure and an ultra-reliable/available low-latency link such as a self-driving vehicle. A level of reliability and latency is essential to control a smart grid, automatize industry, achieve robotics, and control and adjust a drone.

5G is a means of providing streaming evaluated as a few hundred megabits per second to gigabits per second and may complement fiber-to-the-home (FTTH) and cable-based broadband (or DOCSIS). Such fast speed is needed to deliver TV in resolution of 4K or more (6K, 8K, and more), as well as virtual reality and augmented reality. Virtual reality (VR) and augmented reality (AR) applications include almost immersive sports games. A specific application program may require a special network configuration. For example, for VR games, gaming companies need to incorporate a core server into an edge network server of a network operator in order to minimize latency.

Automotive is expected to be a new important motivated force in 5G together with many use cases for mobile communication for vehicles. For example, entertainment for passengers requires high simultaneous capacity and mobile broadband with high mobility. This is because future users continue to expect connection of high quality regardless of their locations and speeds. Another use case of an automotive field is an AR dashboard. The AR dashboard causes a driver to identify an object in the dark in addition to an object seen from a front window and displays a distance from the object and a movement of the object by overlapping information talking to the driver. In the future, a wireless module enables communication between vehicles, information exchange between a vehicle and supporting infrastructure, and information exchange between a vehicle and other connected devices (e.g., devices accompanied by a pedestrian). A safety system guides alternative courses of a behavior so that a driver may drive more safely drive, thereby lowering the danger of an accident. The next stage will be a remotely controlled or self-driven vehicle. This requires very high reliability and very fast communication between different self-driven vehicles and between a vehicle and infrastructure. In the future, a self-driven vehicle will perform all driving activities and a driver will focus only upon abnormal traffic that the vehicle cannot identify. Technical requirements of a self-driven vehicle demand ultra-low latency and ultra-high reliability so that traffic safety is increased to a level that cannot be achieved by human being.

A smart city and a smart home/building mentioned as a smart society will be embedded in a high-density wireless sensor network. A distributed network of an intelligent sensor will identify conditions for costs and energy-efficient maintenance of a city or a home. Similar configurations may be performed for respective households. All of temperature sensors, window and heating controllers, burglar alarms, and home appliances are wirelessly connected. Many of these

sensors are typically low in data transmission rate, power, and cost. However, real-time HD video may be demanded by a specific type of device to perform monitoring.

Consumption and distribution of energy including heat or gas is distributed at a higher level so that automated control of the distribution sensor network is demanded. The smart grid collects information and connects the sensors to each other using digital information and communication technology so as to act according to the collected information. Since this information may include behaviors of a supply company and a consumer, the smart grid may improve distribution of fuels such as electricity by a method having efficiency, reliability, economic feasibility, production sustainability, and automation. The smart grid may also be regarded as another sensor network having low latency.

Mission critical application (e.g. e-health) is one of 5G use scenarios. A health part contains many application programs capable of enjoying benefit of mobile communication. A communication system may support remote treatment that provides clinical treatment in a faraway place. Remote treatment may aid in reducing a barrier against distance and improve access to medical services that cannot be continuously available in a faraway rural area. Remote treatment is also used to perform important treatment and save lives in an emergency situation. The wireless sensor network based on mobile communication may provide remote monitoring and sensors for parameters such as heart rate and blood pressure.

Wireless and mobile communication gradually becomes important in the field of an industrial application. Wiring is high in installation and maintenance cost. Therefore, a possibility of replacing a cable with reconstructible wireless links is an attractive opportunity in many industrial fields. However, in order to achieve this replacement, it is necessary for wireless connection to be established with latency, reliability, and capacity similar to those of the cable and management of wireless connection needs to be simplified. Low latency and a very low error probability are new requirements when connection to 5G is needed.

Logistics and freight tracking are important use cases for mobile communication that enables inventory and package tracking anywhere using a location-based information system. The use cases of logistics and freight typically demand low data rate but require location information with a wide range and reliability.

Referring to FIG. 1, the communication system 1 includes wireless devices, base stations (BSs), and a network. Although FIG. 1 illustrates a 5G network as an example of the network of the communication system 1, the implementations of the present disclosure are not limited to the 5G system, and can be applied to the future communication system beyond the 5G system.

The BSs and the network may be implemented as wireless devices and a specific wireless device 200a may operate as a BS/network node with respect to other wireless devices.

The wireless devices represent devices performing communication using radio access technology (RAT) (e.g., 5G New RAT (NR)) or Long-Term Evolution (LTE)) and may be referred to as communication/radio/5G devices. The wireless devices may include, without being limited to, a robot 100a, vehicles 100b-1 and 100b-2, an eXtended Reality (XR) device 100c, a hand-held device 100d, a home appliance 100e, an Internet of Things (IoT) device 100f, and an Artificial Intelligence (AI) device/server 400. For example, the vehicles may include a vehicle having a wireless communication function, an autonomous driving vehicle, and a vehicle capable of performing communication

between vehicles. The vehicles may include an Unmanned Aerial Vehicle (UAV) (e.g., a drone). The XR device may include an Augmented Reality (AR)/Virtual Reality (VR)/Mixed Reality (MR) device and may be implemented in the form of a Head-Mounted Device (HMD), a Head-Up Display (HUD) mounted in a vehicle, a television, a smartphone, a computer, a wearable device, a home appliance device, a digital signage, a vehicle, a robot, etc. The handheld device may include a smartphone, a smartpad, a wearable device (e.g., a smartwatch or a smartglasses), and a computer (e.g., a notebook). The home appliance may include a TV, a refrigerator, and a washing machine. The IoT device may include a sensor and a smartmeter.

In the present disclosure, the wireless devices **100a** to **100f** may be called user equipments (UEs). A user equipment (UE) may include, for example, a cellular phone, a smartphone, a laptop computer, a digital broadcast terminal, a personal digital assistant (PDA), a portable multimedia player (PMP), a navigation system, a slate personal computer (PC), a tablet PC, an ultrabook, a vehicle, a vehicle having an autonomous traveling function, a connected car, an unmanned aerial vehicle (UAV), an artificial intelligence (AI) module, a robot, an augmented reality (AR) device, a virtual reality (VR) device, a mixed reality (MR) device, a hologram device, a public safety device, an MTC device, an IoT device, a medical device, a FinTech device (or a financial device), a security device, a weather/environment device, a device related to a 5G service, or a device related to a fourth industrial revolution field. The unmanned aerial vehicle (UAV) may be, for example, an aircraft aviated by a wireless control signal without a human being onboard. The VR device may include, for example, a device for implementing an objector a background of the virtual world. The AR device may include, for example, a device implemented by connecting an object or a background of the virtual world to an object or a background of the real world. The MR device may include, for example, a device implemented by merging an objector a background of the virtual world into an object or a background of the real world. The hologram device may include, for example, a device for implementing a stereoscopic image of 360 degrees by recording and reproducing stereoscopic information, using an interference phenomenon of light generated when two laser lights called holography meet. The public safety device may include, for example, an image relay device or an image device that is wearable on the body of a user. The MTC device and the IoT device may be, for example, devices that do not require direct human intervention or manipulation. For example, the MTC device and the IoT device may include smartmeters, vending machines, thermometers, smartbulbs, door locks, or various sensors. The medical device may be, for example, a device used for the purpose of diagnosing, treating, relieving, curing, or preventing disease. For example, the medical device may be a device used for the purpose of diagnosing, treating, relieving, or correcting injury or impairment. For example, the medical device may be a device used for the purpose of inspecting, replacing, or modifying a structure or a function. For example, the medical device may be a device used for the purpose of adjusting pregnancy. For example, the medical device may include a device for treatment, a device for operation, a device for (in vitro) diagnosis, a hearing aid, or a device for procedure. The security device may be, for example, a device installed to prevent a danger that may arise and to maintain safety. For example, the security device may be a camera, a CCTV, a recorder, or a black box. The FinTech device may be, for example, a device capable of

providing a financial service such as mobile payment. For example, the FinTech device may include a payment device or a point of sales (POS) system. The weather/environment device may include, for example, a device for monitoring or predicting a weather/environment.

The wireless devices **100a** to **100f** may be connected to the network **300** via the BSs **200**. An AI technology may be applied to the wireless devices **100a** to **100f** and the wireless devices **100a** to **100f** may be connected to the AI server **400** via the network **300**. The network **300** may be configured using a 3G network, a 4G (e.g., LTE) network, a 5G (e.g., NR) network, and a beyond-5G network. Although the wireless devices **100a** to **100f** may communicate with each other through the BSs **200**/network **300**, the wireless devices **100a** to **100f** may perform direct communication (e.g., sidelink communication) with each other without passing through the BSs/network. For example, the vehicles **100b-1** and **100b-2** may perform direct communication (e.g. Vehicle-to-Vehicle (V2V)/Vehicle-to-everything (V2X) communication). The IoT device (e.g., a sensor) may perform direct communication with other IoT devices (e.g., sensors) or other wireless devices **100a** to **100f**.

Wireless communication/connections **150a** and **150b** may be established between the wireless devices **100a** to **100f**/BS **200**-BS **200**. Herein, the wireless communication/connections may be established through various RATs (e.g., 5G NR) such as uplink/downlink communication **150a** and sidelink communication **150b** (or D2D communication). The wireless devices and the BSs/the wireless devices may transmit/receive radio signals to/from each other through the wireless communication/connections **150a** and **150b**. For example, the wireless communication/connections **150a** and **150b** may transmit/receive signals through various physical channels. To this end, at least a part of various configuration information configuring processes, various signal processing processes (e.g., channel encoding/decoding, modulation/demodulation, and resource mapping/demapping), and resource allocating processes, for transmitting/receiving radio signals, may be performed based on the various proposals of the present disclosure.

FIG. 2 is a block diagram illustrating examples of communication devices which can perform a method according to the present disclosure.

Referring to FIG. 2, a first wireless device **100** and a second wireless device **200** may transmit/receive radio signals to/from an external device through a variety of RATs (e.g., LTE and NR). In FIG. 2, {the first wireless device **100** and the second wireless device **200**} may correspond to {the wireless device **100a** to **100f** and the BS **200**} and/or {the wireless device **100a** to **100f** and the wireless device **100a** to **100f**} of FIG. 1.

The first wireless device **100** may include one or more processors **102** and one or more memories **104** and additionally further include one or more transceivers **106** and/or one or more antennas **108**. The processor(s) **102** may control the memory(s) **104** and/or the transceiver(s) **106** and may be configured to implement the functions, procedures, and/or methods described in the present disclosure. For example, the processor(s) **102** may process information within the memory(s) **104** to generate first information/signals and then transmit radio signals including the first information/signals through the transceiver(s) **106**. The processor(s) **102** may receive radio signals including second information/signals through the transceiver **106** and then store information obtained by processing the second information/signals in the memory(s) **104**. The memory(s) **104** may be connected to the processor(s) **102** and may store a variety of information

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related to operations of the processor(s) 102. For example, the memory(s) 104 may store software code including commands for performing a part or the entirety of processes controlled by the processor(s) 102 or for performing the procedures and/or methods described in the present disclosure. Herein, the processor(s) 102 and the memory(s) 104 may be a part of a communication modem/circuit/chip designed to implement RAT (e.g., LTE or NR). The transceiver(s) 106 may be connected to the processor(s) 102 and transmit and/or receive radio signals through one or more antennas 108. Each of the transceiver(s) 106 may include a transmitter and/or a receiver. The transceiver(s) 106 may be interchangeably used with radio frequency (RF) unit(s). In the present invention, the wireless device may represent a communication modem/circuit/chip.

The second wireless device 200 may include one or more processors 202 and one or more memories 204 and additionally further include one or more transceivers 206 and/or one or more antennas 208. The processor(s) 202 may control the memory(s) 204 and/or the transceiver(s) 206 and may be configured to implement the functions, procedures, and/or methods described in the present disclosure. For example, the processor(s) 202 may process information within the memory(s) 204 to generate third information/signals and then transmit radio signals including the third information/signals through the transceiver(s) 206. The processor(s) 202 may receive radio signals including fourth information/signals through the transceiver(s) 106 and then store information obtained by processing the fourth information/signals in the memory(s) 204. The memory(s) 204 may be connected to the processor(s) 202 and may store a variety of information related to operations of the processor(s) 202. For example, the memory(s) 204 may store software code including commands for performing apart or the entirety of processes controlled by the processor(s) 202 or for performing the procedures and/or methods described in the present disclosure. Herein, the processor(s) 202 and the memory(s) 204 may be apart of a communication modem/circuit/chip designed to implement RAT (e.g., LTE or NR). The transceiver(s) 206 may be connected to the processor(s) 202 and transmit and/or receive radio signals through one or more antennas 208. Each of the transceiver(s) 206 may include a transmitter and/or a receiver. The transceiver(s) 206 may be interchangeably used with RF unit(s). In the present invention, the wireless device may represent a communication modem/circuit/chip.

Hereinafter, hardware elements of the wireless devices 100 and 200 will be described more specifically. One or more protocol layers may be implemented by, without being limited to, one or more processors 102 and 202. For example, the one or more processors 102 and 202 may implement one or more layers (e.g., functional layers such as PHY, MAC, RLC, PDCP, RRC, and SDAP). The one or more processors 102 and 202 may generate one or more Protocol Data Units (PDUs) and/or one or more Service Data Unit (SDUs) according to the functions, procedures, proposals, and/or methods disclosed in the present disclosure. The one or more processors 102 and 202 may generate messages, control information, data, or information according to the functions, procedures, proposals, and/or methods disclosed in the present disclosure and provide the generated signals to the one or more transceivers 106 and 206. The one or more processors

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102 and 202 may receive the signals (e.g., baseband signals) from the one or more transceivers 106 and 206 and acquire the PDUs, SDUs, messages, control information, data, or information according to the functions, procedures, proposals, and/or methods disclosed in the present disclosure.

The one or more processors 102 and 202 may be referred to as controllers, microcontrollers, microprocessors, or microcomputers. The one or more processors 102 and 202 may be implemented by hardware, firmware, software, or a combination thereof. As an example, one or more Application Specific Integrated Circuits (ASICs), one or more Digital Signal Processors (DSPs), one or more Digital Signal Processing Devices (DSPDs), one or more Programmable Logic Devices (PLDs), or one or more Field Programmable Gate Arrays (FPGAs) may be included in the one or more processors 102 and 202. The functions, procedures, proposals, and/or methods disclosed in the present disclosure may be implemented using firmware or software and the firmware or software may be configured to include the modules, procedures, or functions. Firmware or software configured to perform the functions, procedures, proposals, and/or methods disclosed in the present disclosure may be included in the one or more processors 102 and 202 or stored in the one or more memories 104 and 204 so as to be driven by the one or more processors 102 and 202. The functions, procedures, proposals, and/or methods disclosed in the present disclosure may be implemented using firmware or software in the form of code, commands, and/or a set of commands.

The one or more memories 104 and 204 may be connected to the one or more processors 102 and 202 and store various types of data, signals, messages, information, programs, code, instructions, and/or commands. The one or more memories 104 and 204 may be configured by Read-Only Memories (ROMs), Random Access Memories (RAMs), Electrically Erasable Programmable Read-Only Memories (EPROMs), flash memories, hard drives, registers, cash memories, computer-readable storage media, and/or combinations thereof. The one or more memories 104 and 204 may be located at the interior and/or exterior of the one or more processors 102 and 202. The one or more memories 104 and 204 may be connected to the one or more processors 102 and 202 through various technologies such as wired or wireless connection.

The one or more transceivers 106 and 206 may transmit user data, control information, and/or radio signals/channels, mentioned in the methods and/or operational flowcharts of the present disclosure, to one or more other devices. The one or more transceivers 106 and 206 may receive user data, control information, and/or radio signals/channels, mentioned in the functions, procedures, proposals, methods, and/or operational flowcharts disclosed in the present disclosure, from one or more other devices. For example, the one or more transceivers 106 and 206 may be connected to the one or more processors 102 and 202 and transmit and receive radio signals. For example, the one or more processors 102 and 202 may perform control so that the one or more transceivers 106 and 206 may transmit user data, control information, or radio signals to one or more other devices. The one or more processors 102 and 202 may perform control so that the one or more transceivers 106 and 206 may receive user data, control information, or radio signals from one or more other devices. The one or more transceivers 106 and 206 may be connected to the one or more antennas 108 and 208 and the one or more transceivers 106 and 206 may be configured to transmit and receive user data, control information, and/or radio signals/channels,

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mentioned in the functions, procedures, proposals, methods, and/or operational flowcharts disclosed in the present disclosure, through the one or more antennas **108** and **208**. In the present disclosure, the one or more antennas may be a plurality of physical antennas or a plurality of logical antennas (e.g., antenna ports). The one or more transceivers **106** and **206** may convert received radio signals/channels etc. from RF band signals into baseband signals in order to process received user data, control information, radio signals/channels, etc. using the one or more processors **102** and **202**. The one or more transceivers **106** and **206** may convert the user data, control information, radio signals/channels, etc. processed using the one or more processors **102** and **202** from the base band signals into the RF band signals. To this end, the one or more transceivers **106** and **206** may include (analog) oscillators and/or filters. For example, the transceivers **106** and **206** can up-convert OFDM baseband signals to a carrier frequency by their (analog) oscillators and/or filters under the control of the processors **102** and **202** and transmit the up-converted OFDM signals at the carrier frequency. The transceivers **106** and **206** may receive OFDM signals at a carrier frequency and down-convert the OFDM signals into OFDM baseband signals by their (analog) oscillators and/or filters under the control of the transceivers **102** and **202**.

In the implementations of the present disclosure, a UE may operate as a transmitting device in uplink (UL) and as a receiving device in downlink (DL). In the implementations of the present disclosure, a BS may operate as a receiving device in UL and as a transmitting device in DL. Hereinafter, for convenience of description, it is mainly assumed that the first wireless device **100** acts as the UE, and the second wireless device **200** acts as the BS, unless otherwise mentioned or described. For example, the processor(s) **102** connected to, mounted on or launched in the first wireless device **100** may be configured to perform the UE behaviour according to an implementation of the present disclosure or control the transceiver(s) **106** to perform the UE behaviour according to an implementation of the present disclosure. The processor(s) **202** connected to, mounted on or launched in the second wireless device **200** may be configured to perform the BS behaviour according to an implementation of the present disclosure or control the transceiver(s) **206** to perform the BS behaviour according to an implementation of the present disclosure.

FIG. 3 illustrates another example of a wireless device which can perform implementations of the present invention. The wireless device may be implemented in various forms according to a use-case/service (refer to FIG. 1).

Referring to FIG. 3, wireless devices **100** and **200** may correspond to the wireless devices **100** and **200** of FIG. 2 and may be configured by various elements, components, units/portions, and/or modules. For example, each of the wireless devices **100** and **200** may include a communication unit **110**, a control unit **120**, a memory unit **130**, and additional components **140**. The communication unit may include a communication circuit **112** and transceiver(s) **114**. For example, the communication circuit **112** may include the one or more processors **102** and **202** of FIG. 2 and/or the one or more memories **104** and **204** of FIG. 2. For example, the transceiver(s) **114** may include the one or more transceivers **106** and **206** of FIG. 2 and/or the one or more antennas **108** and **208** of FIG. 2. The control unit **120** is electrically connected to the communication unit **110**, the memory **130**, and the additional components **140** and controls overall operation of the wireless devices. For example, the control unit **120** may control an electric/mechanical operation of the

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wireless device based on programs/code/commands/information stored in the memory unit **130**. The control unit **120** may transmit the information stored in the memory unit **130** to the exterior (e.g., other communication devices) via the communication unit **110** through a wireless/wired interface or store, in the memory unit **130**, information received through the wireless/wired interface from the exterior (e.g., other communication devices) via the communication unit **110**.

The additional components **140** may be variously configured according to types of wireless devices. For example, the additional components **140** may include at least one of a power unit/battery, input/output (I/O) unit (e.g. audio I/O port, video I/O port), a driving unit, and a computing unit. The wireless device may be implemented in the form of, without being limited to, the robot (**100a** of FIG. 1), the vehicles (**100b-1** and **100b-2** of FIG. 1), the XR device (**100c** of FIG. 1), the hand-held device (**100d** of FIG. 1), the home appliance (**100e** of FIG. 1), the IoT device (**100f** of FIG. 1), a digital broadcast terminal, a hologram device, a public safety device, an MTC device, a medicine device, a Fintech device (or a finance device), a security device, a climate/environment device, the AI server/device (**400** of FIG. 1), the BSs (**200** of FIG. 1), a network node, etc. The wireless device may be used in a mobile or fixed place according to a use-example/service.

In FIG. 3, the entirety of the various elements, components, units/portions, and/or modules in the wireless devices **100** and **200** may be connected to each other through a wired interface or at least a part thereof may be wirelessly connected through the communication unit **110**. For example, in each of the wireless devices **100** and **200**, the control unit **120** and the communication unit **110** may be connected by wire and the control unit **120** and first units (e.g., **130** and **140**) may be wirelessly connected through the communication unit **110**. Each element, component, unit/portion, and/or module within the wireless devices **100** and **200** may further include one or more elements. For example, the control unit **120** may be configured by a set of one or more processors. As an example, the control unit **120** may be configured by a set of a communication control processor, an application processor, an electronic control unit (ECU), a graphical processing unit, and a memory control processor. As another example, the memory **130** may be configured by a random access memory (RAM), a dynamic RAM (DRAM), a read only memory (ROM), a flash memory, a volatile memory, a non-volatile memory, and/or a combination thereof.

FIG. 4 illustrates an example of protocol stacks in a 3GPP based wireless communication system.

In particular, FIG. 4(a) illustrates an example of a radio interface user plane protocol stack between a UE and a base station (BS) and FIG. 4(b) illustrates an example of a radio interface control plane protocol stack between a UE and a BS. The control plane refers to a path through which control messages used to manage call by a UE and a network are transported. The user plane refers to a path through which data generated in an application layer, for example, voice data or Internet packet data are transported. Referring to FIG. 4(a), the user plane protocol stack may be divided into a first layer (Layer 1) (i.e., a physical (PHY) layer) and a second layer (Layer 2). Referring to FIG. 4(b), the control plane protocol stack may be divided into Layer 1 (i.e., a PHY layer), Layer 2, Layer 3 (e.g., a radio resource control (RRC) layer), and a non-access stratum (NAS) layer. Layer 1, Layer 2 and Layer 3 are referred to as an access stratum (AS).

The NAS control protocol is terminated in an access management function (AMF) on the network side, and performs functions such as authentication, mobility management, security control and etc.

In the 3GPP LTE system, the layer 2 is split into the following sublayers: medium access control (MAC), radio link control (RLC), and packet data convergence protocol (PDCP). In the 3GPP New Radio (NR) system, the layer 2 is split into the following sublayers: MAC, RLC, PDCP and SDAP. The PHY layer offers to the MAC sublayer transport channels, the MAC sublayer offers to the RLC sublayer logical channels, the RLC sublayer offers to the PDCP sublayer RLC channels, the PDCP sublayer offers to the SDAP sublayer radio bearers. The SDAP sublayer offers to 5G Core Network quality of service (QoS) flows.

In the 3GPP NR system, the main services and functions of SDAP include: mapping between a QoS flow and a data radio bearer; marking QoS flow ID (QFI) in both DL and UL packets. A single protocol entity of SDAP is configured for each individual PDU session.

In the 3GPP NR system, the main services and functions of the RRC sublayer include: broadcast of system information related to AS and NAS; paging initiated by 5G core (5GC) or NG-RAN; establishment, maintenance and release of an RRC connection between the UE and NG-RAN; security functions including key management; establishment, configuration, maintenance and release of signalling radio bearers (SRBs) and data radio bearers (DRBs); mobility functions (including: handover and context transfer; UE cell selection and reselection and control of cell selection and reselection; Inter-RAT mobility); QoS management functions; UE measurement reporting and control of the reporting; detection of and recovery from radio link failure; NAS message transfer to/from NAS from/to UE.

In the 3GPP NR system, the main services and functions of the PDCP sublayer for the user plane include: sequence numbering; header compression and decompression: ROHC only; transfer of user data; reordering and duplicate detection; in-order delivery; PDCP PDU routing (in case of split bearers); retransmission of PDCP SDUs; ciphering, deciphering and integrity protection; PDCP SDU discard; PDCP re-establishment and data recovery for RLC AM; PDCP status reporting for RLC AM; duplication of PDCP PDUs and duplicate discard indication to lower layers. The main services and functions of the PDCP sublayer for the control plane include: sequence numbering; ciphering, deciphering and integrity protection; transfer of control plane data; reordering and duplicate detection; in-order delivery; duplication of PDCP PDUs and duplicate discard indication to lower layers.

The RLC sublayer supports three transmission modes: Transparent Mode (TM); Un-acknowledged Mode (UM); and Acknowledged Mode (AM). The RLC configuration is per logical channel with no dependency on numerologies and/or transmission durations. In the 3GPP NR system, the main services and functions of the RLC sublayer depend on the transmission mode and include: Transfer of upper layer PDUs; sequence numbering independent of the one in PDCP (UM and AM); error correction through ARQ (AM only); segmentation (AM and UM) and re-segmentation (AM only) of RLC SDUs; reassembly of SDU (AM and UM); duplicate detection (AM only); RLC SDU discard (AM and UM); RLC re-establishment; protocol error detection (AM only).

In the 3GPP NR system, the main services and functions of the MAC sublayer include: mapping between logical channels and transport channels; multiplexing/de-multiplexing of MAC SDUs belonging to one or different logical

channels into/from transport blocks (TB) delivered to/from the physical layer on transport channels; scheduling information reporting; error correction through HARQ (one HARQ entity per cell in case of carrier aggregation (CA)); priority handling between UEs by means of dynamic scheduling; priority handling between logical channels of one UE by means of logical channel prioritization; padding. A single MAC entity may support multiple numerologies, transmission timings and cells. Mapping restrictions in logical channel prioritization control which numerology(ies), cell(s), and transmission timing(s) a logical channel can use. Different kinds of data transfer services are offered by MAC. To accommodate different kinds of data transfer services, multiple types of logical channels are defined i.e. each supporting transfer of a particular type of information. Each logical channel type is defined by what type of information is transferred. Logical channels are classified into two groups: Control Channels and Traffic Channels. Control channels are used for the transfer of control plane information only, and traffic channels are used for the transfer of user plane information only. Broadcast Control Channel (BCCH) is a downlink logical channel for broadcasting system control information, paging Control Channel (PCCH) is a downlink logical channel that transfers paging information, system information change notifications and indications of ongoing PWS broadcasts, Common Control Channel (CCCH) is a logical channel for transmitting control information between UEs and network and used for UEs having no RRC connection with the network, and Dedicated Control Channel (DCCH) is a point-to-point bi-directional logical channel that transmits dedicated control information between a UE and the network and used by UEs having an RRC connection. Dedicated Traffic Channel (DTCH) is a point-to-point logical channel, dedicated to one UE, for the transfer of user information. ADTCH can exist in both uplink and downlink. In Downlink, the following connections between logical channels and transport channels exist: BCCH can be mapped to BCH; BCCH can be mapped to downlink shared channel (DL-SCH); PCCH can be mapped to PCH; CCCH can be mapped to DL-SCH; DCCH can be mapped to DL-SCH; and DTCH can be mapped to DL-SCH. In Uplink, the following connections between logical channels and transport channels exist: CCCH can be mapped to uplink shared channel (UL-SCH); DCCH can be mapped to UL-SCH; and DTCH can be mapped to UL-SCH.

FIG. 5 illustrates an example of a frame structure in a 3GPP based wireless communication system.

The frame structure illustrated in FIG. 5 is purely exemplary and the number of subframes, the number of slots, and/or the number of symbols in a frame may be variously changed. In the 3GPP based wireless communication system, OFDM numerologies (e.g., subcarrier spacing (SCS), transmission time interval (TTI) duration) may be differently configured between a plurality of cells aggregated for one UE. For example, if a UE is configured with different SCSs for cells aggregated for the cell, an (absolute time) duration of a time resource (e.g. a subframe, a slot, or a TTI) including the same number of symbols may be different among the aggregated cells. Herein, symbols may include OFDM symbols (or CP-OFDM symbols), SC-FDMA symbols (or discrete Fourier transform-spread-OFDM (DFT-s-OFDM) symbols).

Referring to FIG. 5, downlink and uplink transmissions are organized into frames. Each frame has $T_f=10$ ms duration. Each frame is divided into two half-frames, where each of the half-frames has 5 ms duration. Each half-frame consists of 5 subframes, where the duration T_{sf} per subframe

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is 1 ms. Each subframe is divided into slots and the number of slots in a subframe depends on a subcarrier spacing. Each slot includes 14 or 12 OFDM symbols based on a cyclic prefix (CP). In a normal CP, each slot includes 14 OFDM symbols and, in an extended CP, each slot includes 12 OFDM symbols. The numerology is based on exponentially scalable subcarrier spacing $\Delta f = 2^u \cdot 15$ kHz. The following table shows the number of OFDM symbols per slot, the number of slots per frame, and the number of slots per for the normal CP, according to the subcarrier spacing $\Delta f = 2^u \cdot 15$ kHz.

TABLE 1

u	$N_{\text{symbol}}^{\text{slot}}$	$N_{\text{slot}}^{\text{frame}, u}$	$N_{\text{slot}}^{\text{subframe}, u}$
0	14	10	1
1	14	20	2
2	14	40	4
3	14	80	8
4	14	160	16

The following table shows the number of OFDM symbols per slot, the number of slots per frame, and the number of slots per for the extended CP, according to the subcarrier spacing $\Delta f = 2^u \cdot 15$ kHz.

TABLE 2

u	$N_{\text{symbol}}^{\text{slot}}$	$N_{\text{slot}}^{\text{frame}, u}$	$N_{\text{slot}}^{\text{subframe}, u}$
2	12	40	4

A slot includes plural symbols (e.g., 14 or 12 symbols) in the time domain. For each numerology (e.g. subcarrier spacing) and carrier, a resource grid of $N_{\text{symbol}}^{\text{size}, u} \cdot N_{\text{sc}}^{\text{RB}}$ subcarriers and $N_{\text{symbol}}^{\text{subframe}, u}$ OFDM symbols is defined, starting at common resource block (CRB) $N_{\text{start}, u}^{\text{grid}}$ indicated by higher-layer signaling (e.g. radio resource control (RRC) signaling), where $N_{\text{symbol}}^{\text{size}, u}$ is the number of resource blocks in the resource grid and the subscript x is DL for downlink and UL for uplink. $N_{\text{sc}}^{\text{RB}}$ is the number of subcarriers per resource blocks. In the 3GPP based wireless communication system, $N_{\text{sc}}^{\text{RB}}$ is 12 generally. There is one resource grid for a given antenna port p, subcarrier spacing configuration u, and transmission direction (DL or UL). The carrier bandwidth $N_{\text{symbol}}^{\text{size}, u}$ for subcarrier spacing configuration u is given by the higher-layer parameter (e.g. RRC parameter). Each element in the resource grid for the antenna port p and the subcarrier spacing configuration u is referred to as a resource element (RE) and one complex symbol may be mapped to each RE. Each RE in the resource grid is uniquely identified by an index k in the frequency domain and an index l representing a symbol location relative to a reference point in the time domain. In the 3GPP based wireless communication system, a resource block is defined by 12 consecutive subcarriers in the frequency domain.

In the 3GPP NR system, resource blocks are classified into CRBs and physical resource blocks (PRBs). CRBs are numbered from 0 and upwards in the frequency domain for subcarrier spacing configuration u. The center of subcarrier 0 of CRB 0 for subcarrier spacing configuration u coincides with 'point A' which serves as a common reference point for resource block grids. In the 3GPP NR system, PRBs are defined within a bandwidth part (BWP) and numbered from 0 to $N_{\text{sizeBWP}, i}^{\text{sizeBWP}, i} - 1$, where i is the number of the bandwidth part. The relation between the physical resource block n_{PRB}

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in the bandwidth part i and the common resource block n_{CRB} is as follows: $n_{\text{PRB}} = n_{\text{CRB}} + N_{\text{sizeBWP}, i}^{\text{sizeBWP}, i}$, where $N_{\text{sizeBWP}, i}^{\text{sizeBWP}, i}$ is the common resource block where bandwidth part starts relative to CRB 0. The BWP includes a plurality of consecutive resource blocks. A carrier may include a maximum of N (e.g., 5) BWPs. A UE may be configured with one or more BWPs on a given component carrier. Only one BWP among BWPs configured to the UE can active at a time. The active BWP defines the UE's operating bandwidth within the cell's operating bandwidth.

NR frequency bands are defined as 2 types of frequency range, FR1 and FR2. FR2 is may also called millimeter wave (mmW). The frequency ranges in which NR can operate are identified as described in Table 3.

TABLE 3

Frequency Range designation	Corresponding frequency range	Subcarrier Spacing
FR1	450 MHz-7125 MHz	15, 30, 60 kHz
FR2	24250 MHz-52600 MHz	60, 120, 240 kHz

FIG. 6 illustrates a data flow example in the 3GPP NR system.

In FIG. 6, "RB" denotes a radio bearer, and "H" denotes a header. Radio bearers are categorized into two groups: data radio bearers (DRB) for user plane data and signalling radio bearers (SRB) for control plane data. The MAC PDU is transmitted/received using radio resources through the PHY layer to/from an external device. The MAC PDU arrives to the PHY layer in the form of a transport block.

In the PHY layer, the uplink transport channels UL-SCH and RACH are mapped to physical uplink shared channel (PUSCH) and physical random access channel (PRACH), respectively, and the downlink transport channels DL-SCH, BCH and PCH are mapped to physical downlink shared channel (PDSCH), physical broadcast channel (PBCH) and PDSCH, respectively. In the PHY layer, uplink control information (UCI) is mapped to PUCCH, and downlink control information (DCI) is mapped to PDCCH. A MAC PDU related to UL-SCH is transmitted by a UE via a PUSCH based on an UL grant, and a MAC PDU related to DL-SCH is transmitted by a BS via a PDSCH based on a DL assignment.

In order to transmit data unit(s) of the present disclosure on UL-SCH, a UE shall have uplink resources available to the UE. In order to receive data unit(s) of the present disclosure on DL-SCH, a UE shall have downlink resources available to the UE. The resource allocation includes time domain resource allocation and frequency domain resource allocation. In the present disclosure, uplink resource allocation is also referred to as uplink grant, and downlink resource allocation is also referred to as downlink assignment. An uplink grant is either received by the UE dynamically on PDCCH, in a Random Access Response, or configured to the UE semi-persistently by RRC. Downlink assignment is either received by the UE dynamically on the PDCCH, or configured to the UE semi-persistently by RRC signaling from the BS.

In UL, the BS can dynamically allocate resources to UEs via the Cell Radio Network Temporary Identifier (C-RNTI) on PDCCH(s). A UE always monitors the PDCCH(s) in order to find possible grants for uplink transmission when its downlink reception is enabled (activity governed by discontinuous reception (DRX) when configured). In addition, with Configured Grants, the BS can allocate uplink resources for

the initial HARQ transmissions to UEs. Two types of configured uplink grants are defined: Type 1 and Type 2. With Type 1, RRC directly provides the configured uplink grant (including the periodicity). With Type 2, RRC defines the periodicity of the configured uplink grant while PDCCH addressed to Configured Scheduling RNTI (CS-RNTI) can either signal and activate the configured uplink grant, or deactivate it; i.e. a PDCCH addressed to CS-RNTI indicates that the uplink grant can be implicitly reused according to the periodicity defined by RRC, until deactivated.

In DL, the BS can dynamically allocate resources to UEs via the C-RNTI on PDCCH(s). A UE always monitors the PDCCH(s) in order to find possible assignments when its downlink reception is enabled (activity governed by DRX when configured). In addition, with Semi-Persistent Scheduling (SPS), the BS can allocate downlink resources for the initial HARQ transmissions to UEs: RRC defines the periodicity of the configured downlink assignments while PDCCH addressed to CS-RNTI can either signal and activate the configured downlink assignment, or deactivate it. In other words, a PDCCH addressed to CS-RNTI indicates that the downlink assignment can be implicitly reused according to the periodicity defined by RRC, until deactivated.

<Resource Allocation by PDCCH (i.e. Resource Allocation by DCI)>

PDCCH can be used to schedule DL transmissions on PDSCH and UL transmissions on PUSCH, where the downlink control information (DCI) on PDCCH includes: downlink assignments containing at least modulation and coding format (e.g., modulation and coding scheme (MCS) index IMCS), resource allocation, and hybrid-ARQ information related to DL-SCH; or uplink scheduling grants containing at least modulation and coding format, resource allocation, and hybrid-ARQ information related to UL-SCH. The size and usage of the DCI carried by one PDCCH are varied depending on DCI formats. For example, in the 3GPP NR system, DCI format 0_0 or DCI format 0_1 is used for scheduling of PUSCH in one cell, and DCI format 1_0 or DCI format 1_1 is used for scheduling of PDSCH in one cell.

FIG. 7 illustrates an example of PDSCH time domain resource allocation by PDCCH, and an example of PUSCH time domain resource allocation by PDCCH.

Downlink control information (DCI) carried by a PDCCH for scheduling PDSCH or PUSCH includes a value m for a row index $m+1$ to an allocation table for PDSCH or PUSCH. Either a predefined default PDSCH time domain allocation A, B or C is applied as the allocation table for PDSCH, or RRC configured `pdsch-TimeDomainAllocationList` is applied as the allocation table for PDSCH. Either a predefined default PUSCH time domain allocation A is applied as the allocation table for PUSCH, or the RRC configured `pusch-TimeDomainAllocationList` is applied as the allocation table for PUSCH. Which PDSCH time domain resource allocation configuration to apply and which PUSCH time domain resource allocation table to apply are determined according to a fixed/predefined rule (e.g. Table 5.1.2.1.1-1 in 3GPP TS 38.214v15.3.0, Table 6.1.2.1.1-1 in 3GPP TS 38.214v15.3.0).

Each indexed row in PDSCH time domain allocation configurations defines the slot offset K_0 , the start and length indicator SLIV, or directly the start symbol S and the allocation length L , and the PDSCH mapping type to be assumed in the PDSCH reception. Each indexed row in PUSCH time domain allocation configurations defines the slot offset K_2 , the start and length indicator SLIV, or directly the start symbol S and the allocation length L , and the

PUSCH mapping type to be assumed in the PUSCH reception. K_0 for PDSCH, or K_2 for PUSCH is the timing difference between a slot with a PDCCH and a slot with PDSCH or PUSCH corresponding to the PDCCH. SLIV is a joint indication of starting symbol S relative to the start of the slot with PDSCH or PUSCH, and the number L of consecutive symbols counting from the symbol S . For PDSCH/PUSCH mapping type, there are two mapping types: one is Mapping Type A where demodulation reference signal (DMRS) is positioned in 3rd or 4th symbol of a slot depending on the RRC signaling, and other one is Mapping Type B where DMRS is positioned in the first allocated symbol.

The scheduling DCI includes the Frequency domain resource assignment field which provides assignment information on resource blocks used for PDSCH or PUSCH. For example, the Frequency domain resource assignment field may provide a UE with information on a cell for PDSCH or PUSCH transmission, information on a bandwidth part for PDSCH or PUSCH transmission, information on resource blocks for PDSCH or PUSCH transmission.

<Resource Allocation by RRC>

As mentioned above, in uplink, there are two types of transmission without dynamic grant: configured grant Type 1 where an uplink grant is provided by RRC, and stored as configured grant; and configured grant Type 2 where an uplink grant is provided by PDCCH, and stored or cleared as configured uplink grant based on L1 signaling indicating configured uplink grant activation or deactivation. Type 1 and Type 2 are configured by RRC per serving cell and per BWP. Multiple configurations can be active simultaneously only on different serving cells. For Type 2, activation and deactivation are independent among the serving cells. For the same serving cell, the MAC entity is configured with either Type 1 or Type 2.

A UE is provided with at least the following parameters via RRC signaling from a BS when the configured grant type 1 is configured:

- `cs-RNTI` which is CS-RNTI for retransmission;
- `periodicity` which provides periodicity of the configured grant Type 1;
- `timeDomainOffset` which represents offset of a resource with respect to $SFN=0$ in time domain;
- `timeDomainAllocation` value m which provides a row index $m+1$ pointing to an allocation table, indicating a combination of a start symbol S and length L and PUSCH mapping type;
- `frequencyDomainAllocation` which provides frequency domain resource allocation; and
- `mcsAndTBS` which provides IMCS representing the modulation order, target code rate and transport block size. Upon configuration of a configured grant Type 1 for a serving cell by RRC, the UE stores the uplink grant provided by RRC as a configured uplink grant for the indicated serving cell, and initialise or re-initialise the configured uplink grant to start in the symbol according to time DomainOffset and S (derived from SLIV), and to reoccur with periodicity. After an uplink grant is configured for a configured grant Type 1, the UE considers that the uplink grant recurs associated with each symbol for which:

$$[(SFN * \text{numberOfSlotsPerFrame} + (\text{slot number in the frame} * \text{numberOfSymbolsPerSlot}) + \text{symbol number in the slot}] = (\text{timeDomainOffset} * \text{numberOfSymbolsPerSlot} + S +$$

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$N \times \text{periodicity}) \bmod (1024 \times \text{numberOfSlotsPerFrame} \times \text{numberOfSymbolsPerSlot})$, for all $N \geq 0$.

A UE is provided with at least the following parameters via RRC signaling from a BS when the configured grant Type 2 is configured:

cs-RNTI which is CS-RNTI for activation, deactivation, and retransmission; and
periodicity which provides periodicity of the configured grant Type 2. The actual uplink grant is provided to the UE by the PDCCH (addressed to CS-RNTI). After an uplink grant is configured for a configured grant Type 2, the UE considers that the uplink grant recurs associated with each symbol for which:

$$[(\text{SFN} \times \text{numberOfSlotsPerFrame} \times \text{numberOfSymbolsPerSlot}) + (\text{slot number in the frame} \times \text{numberOfSymbolsPerSlot}) + \text{symbol number in the slot}] = [(\text{SFN}_{\text{starttime}} \times \text{numberOfSlotsPerFrame} \times \text{numberOfSymbolsPerSlot} + \text{slot}_{\text{starttime}} \times \text{numberOfSymbolsPerSlot} + \text{symbol}_{\text{starttime}}) + N \times \text{periodicity}] \bmod (1024 \times \text{numberOfSlotsPerFrame} \times \text{numberOfSymbolsPerSlot})$$
, for all $N \geq 0$, where $\text{SFN}_{\text{starttime}}$, $\text{slot}_{\text{starttime}}$, and $\text{symbol}_{\text{starttime}}$ are the SFN, slot, and symbol, respectively, of the first transmission opportunity of PUSCH where the configured uplink grant was (re-)initialised. $\text{numberOfSlotsPerFrame}$ and $\text{numberOfSymbolsPerSlot}$ refer to the number of consecutive slots per frame and the number of consecutive OFDM symbols per slot, respectively.

For configured uplink grants, the HARQ Process ID associated with the first symbol of a UL transmission is derived from the following equation:

$$\text{HARQ Process ID} = [\text{floor}(\text{CURRENT_symbol} / \text{periodicity})] \bmod \text{nrofHARQ-Processes}$$

where $\text{CURRENT_symbol} = (\text{SFN} \times \text{numberOfSlotsPerFrame} \times \text{numberOfSymbolsPerSlot} + \text{slot number in the frame} \times \text{numberOfSymbolsPerSlot} + \text{symbol number in the slot})$, and $\text{numberOfSlotsPerFrame}$ and $\text{numberOfSymbolsPerSlot}$ refer to the number of consecutive slots per frame and the number of consecutive symbols per slot, respectively as specified in TS 38.211. CURRENT_symbol refers to the symbol index of the first transmission occasion of a repetition bundle that takes place. A HARQ process is configured for a configured uplink grant if the configured uplink grant is activated and the associated HARQ process ID is less than $\text{nrofHARQ-Processes}$.

For downlink, a UE may be configured with semi-persistent scheduling (SPS) per serving cell and per BWP by RRC signaling from a BS. Multiple configurations can be active simultaneously only on different serving cells. Activation and deactivation of the DL SPS are independent among the serving cells. For DL SPS, a DL assignment is provided to the UE by PDCCH, and stored or cleared based on L1 signaling indicating SPS activation or deactivation. A UE is provided with the following parameters via RRC signaling from a BS when SPS is configured:

cs-RNTI which is CS-RNTI for activation, deactivation, and retransmission;
 $\text{nrofHARQ-Processes}$: which provides the number of configured HARQ processes for SPS;
periodicity which provides periodicity of configured downlink assignment for SPS.

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When SPS is released by upper layers, all the corresponding configurations shall be released.

After a downlink assignment is configured for SPS, the UE considers sequentially that the N^{th} downlink assignment occurs in the slot for which:

$$(\text{numberOfSlotsPerFrame} \times \text{SFN} + \text{slot number in the frame}) = [(\text{numberOfSlotsPerFrame} \times \text{SFN}_{\text{starttime}} + \text{slot}_{\text{starttime}}) + N \times \text{periodicity} \times \text{numberOfSlotsPerFrame} / 10] \bmod (1024 \times \text{numberOfSlotsPerFrame})$$
, where $\text{SFN}_{\text{starttime}}$ and $\text{slot}_{\text{starttime}}$ are the SFN and slot, respectively, of the first transmission of PDSCH where the configured downlink assignment was (re-)initialised.

For configured downlink assignments, the HARQ Process ID associated with the slot where the DL transmission starts is derived from the following equation:

$$\text{HARQ Process ID} = [\text{floor}(\text{CURRENT_slot} \times 10 / (\text{numberOfSlotsPerFrame} \times \text{periodicity}))] \bmod \text{nrofHARQ-Processes}$$

where $\text{CURRENT_slot} = [(\text{SFN} \times \text{numberOfSlotsPerFrame}) + \text{slot number in the frame}]$ and $\text{numberOfSlotsPerFrame}$ refers to the number of consecutive slots per frame as specified in TS 38.211.

A UE validates, for scheduling activation or scheduling release, a DL SPS assignment PDCCH or configured UL grant type 2 PDCCH if the cyclic redundancy check (CRC) of a corresponding DCI format is scrambled with CS-RNTI provided by the RRC parameter cs-RNTI and the new data indicator field for the enabled transport block is set to 0. Validation of the DCI format is achieved if all fields for the DCI format are set according to Table 4 or Table 5. Table 4 shows special fields for DL SPS and UL grant Type 2 scheduling activation PDCCH validation, and Table 5 shows special fields for DL SPS and UL grant Type 2 scheduling release PDCCH validation.

TABLE 4

	DCI format 0_0/0_1	DCI format 1_0	DCI format 1_1
HARQ process number	set to all '0's	set to all '0's	set to all '0's
Redundancy version	set to '00'	set to '00'	For the enabled transport block: set to '00'

TABLE 5

	DCI format 0_0	DCI format 1_0
HARQ process number	set to all '0's	set to all '0's
Redundancy version	set to '00'	set to '00'
Modulation and coding scheme	set to all '1's	set to all '1's
Resource block assignment	set to all '1's	set to all '1's

Actual DL assignment and actual UL grant, and the corresponding modulation and coding scheme are provided by the resource assignment fields (e.g. time domain resource assignment field which provides Time domain resource assignment value m , frequency domain resource assignment field which provides the frequency resource block allocation, modulation and coding scheme field) in the DCI format carried by the DL SPS and UL grant Type 2 scheduling activation PDCCH. If validation is achieved, the UE considers the information in the DCI format as valid activation or valid release of DL SPS or configured UL grant Type 2.

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For UL, the processor(s) 102 of the present disclosure may transmit (or control the transceiver(s) 106 to transmit) the data unit of the present disclosure based on the UL grant available to the UE. The processor(s) 202 of the present disclosure may receive (or control the transceiver(s) 206 to receive) the data unit of the present disclosure based on the UL grant available to the UE.

For DL, the processor(s) 102 of the present disclosure may receive (or control the transceiver(s) 106 to receive) DL data of the present disclosure based on the DL assignment available to the UE. The processor(s) 202 of the present disclosure may transmit (or control the transceiver(s) 206 to transmit) DL data of the present disclosure based on the DL assignment available to the UE.

The data unit(s) of the present disclosure is(are) subject to the physical layer processing at a transmitting side before transmission via radio interface, and the radio signals carrying the data unit(s) of the present disclosure are subject to the physical layer processing at a receiving side. For example, a MAC PDU including the PDCP PDU according to the present disclosure may be subject to the physical layer processing as follows.

FIG. 8 illustrates an example of physical layer processing at a transmitting side.

The following tables show the mapping of the transport channels (TrCHs) and control information to its corresponding physical channels. In particular, Table 6 specifies the mapping of the uplink transport channels to their corresponding physical channels, Table 7 specifies the mapping of the uplink control channel information to its corresponding physical channel, Table 8 specifies the mapping of the downlink transport channels to their corresponding physical channels, and Table 9 specifies the mapping of the downlink control channel information to its corresponding physical channel.

TABLE 6

TrCH	Physical Channel
UL-SCH	PUSCH
RACH	PRACH

TABLE 7

Control information	Physical Channel
UCI	PUCCH, PUSCH

TABLE 8

TrCH	Physical Channel
DL-SCH	PDSCH
BCH	PBCH
PCH	PDSCH

TABLE 9

Control information	Physical Channel
DCI	PDCCH

<Encoding>

Data and control streams from/to MAC layer are encoded to offer transport and control services over the radio trans-

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mission link in the PHY layer. For example, a transport block from MAC layer is encoded into a codeword at a transmitting side. Channel coding scheme is a combination of error detection, error correcting, rate matching, interleaving and transport channel or control information mapping onto/splitting from physical channels.

In the 3GPP NR system, following channel coding schemes are used for the different types of TrCH and the different control information types.

TABLE 10

TrCH	Coding scheme
UL-SCH	LDPC
DL-SCH	
PCH	
BCH	Polar code

TABLE 11

Control Information	Coding scheme
DCI	Polar code
UCI	Block code
	Polar code

For transmission of a DL transport block (i.e. a DL MAC PDU) or a UL transport block (i.e. a UL MAC PDU), a transport block CRC sequence is attached to provide error detection for a receiving side. In the 3GPP NR system, the communication device uses low density parity check (LDPC) codes in encoding/decoding UL-SCH and DL-SCH. The 3GPP NR system supports two LDPC base graphs (i.e. two LDPC base matrixes): LDPC base graph 1 optimized for small transport blocks and LDPC base graph 2 for larger transport blocks. Either LDPC base graph 1 or 2 is selected based on the size of the transport block and coding rate R. The coding rate R is indicated by the modulation coding scheme (MCS) index IMCS. The MCS index is dynamically provided to a UE by PDCCH scheduling PUSCH or PDSCH, provided to a UE by PDCCH activating or (re-) initializing the UL configured grant 2 or DL SPS, or provided to a UE by RRC signaling related to the UL configured grant Type 1. If the CRC attached transport block is larger than the maximum code block size for the selected LDPC base graph, the CRC attached transport block may be segmented into code blocks, and an additional CRC sequence is attached to each code block. The maximum code block sizes for the LDPC base graph 1 and the LDPC base graph 2 are 8448 bits and 3480 bits, respectively. If the CRC attached transport block is not larger than the maximum code block size for the selected LDPC base graph, the CRC attached transport block is encoded with the selected LDPC base graph. Each code block of the transport block is encoded with the selected LDPC base graph. The LDPC coded blocks are then individually rate matched. Code block concatenation is performed to create a codeword for transmission on PDSCH or PUSCH. For PDSCH, up to 2 codewords (i.e. up to 2 transport blocks) can be transmitted simultaneously on the PDSCH. PUSCH can be used for transmission of UL-SCH data and layer 1/2 control information. Although not shown in FIG. 8, the layer 1/2 control information may be multiplexed with the codeword for UL-SCH data.

<Scrambling and Modulation>

The bits of the codeword are scrambled and modulated to generate a block of complex-valued modulation symbols.

<Layer Mapping>

The complex-valued modulation symbols of the codeword are mapped to one or more multiple input multiple output (MIMO) layers. A codeword can be mapped to up to 4 layers. A PDSCH can carry two codewords, and thus a PDSCH can support up to 8-layer transmission. A PUSCH supports a single codeword, and thus a PUSCH can support up to 4-layer transmission.

<Transform Precoding>

The DL transmission waveform is conventional OFDM using a cyclic prefix (CP). For DL, transform precoding (in other words, discrete Fourier transform (DFT)) is not applied.

The UL transmission waveform is conventional OFDM using a CP with a transform precoding function performing DFT spreading that can be disabled or enabled. In the 3GPP NR system, for UL, the transform precoding can be optionally applied if enabled. The transform precoding is to spread UL data in a special way to reduce peak-to-average power ratio (PAPR) of the waveform. The transform precoding is a form of DFT. In other words, the 3GPP NR system supports two options for UL waveform: one is CP-OFDM (same as DL waveform) and the other one is DFT-s-OFDM. Whether a UE has to use CP-OFDM or DFT-s-OFDM is configured by a BS via RRC parameters.

<Subcarrier Mapping>

The layers are mapped to antenna ports. In DL, for the layers to antenna ports mapping, a transparent manner (non-codebook based) mapping is supported and how beamforming or MIMO precoding is performed is transparent to the UE. In UL, for the layers to antenna ports mapping, both the non-codebook based mapping and a codebook based mapping are supported.

For each antenna port (i.e. layer) used for transmission of the physical channel (e.g. PDSCH, PUSCH), the complex-valued modulation symbols are mapped to subcarriers in resource blocks allocated to the physical channel.

<OFDM Modulation>

The communication device at the transmitting side generates a time-continuous OFDM baseband signal on antenna port *p* and subcarrier spacing configuration *u* for OFDM symbol *l* in a TTI for a physical channel by adding a cyclic prefix (CP) and performing IFFT. For example, for each OFDM symbol, the communication device at the transmitting side may perform inverse fast Fourier transform (IFFT) on the complex-valued modulation symbols mapped to resource blocks in the corresponding OFDM symbol and add a CP to the IFFT-ed signal to generate the OFDM baseband signal.

<Up-Conversion>

The communication device at the transmitting side up-converts the OFDM baseband signal for antenna port *p*, subcarrier spacing configuration *u* and OFDM symbol *l* to a carrier frequency *f₀* of a cell to which the physical channel is assigned.

The processors 102 and 202 in FIG. 2 may be configured to perform encoding, scrambling, modulation, layer mapping, transform precoding (for UL), subcarrier mapping, and OFDM modulation. The processors 102 and 202 may control the transceivers 106 and 206 connected to the processors 102 and 202 to up-convert the OFDM baseband signal onto the carrier frequency to generate radio frequency (RF) signals. The radio frequency signals are transmitted through antennas 108 and 208 to an external device.

FIG. 9 illustrates an example of physical layer processing at a receiving side.

The physical layer processing at the receiving side is basically the inverse processing of the physical layer processing at the transmitting side.

<Frequency Down-Conversion>

The communication device at a receiving side receives RF signals at a carrier frequency through antennas. The transceivers 106 and 206 receiving the RF signals at the carrier frequency down-converts the carrier frequency of the RF signals into the baseband in order to obtain OFDM baseband signals.

<OFDM Demodulation>

The communication device at the receiving side obtains complex-valued modulation symbols via CP detachment and FFT. For example, for each OFDM symbol, the communication device at the receiving side removes a CP from the OFDM baseband signals and performs FFT on the CP-removed OFDM baseband signals to obtain complex-valued modulation symbols for antenna port *p*, subcarrier spacing *u* and OFDM symbol *l*.

<Subcarrier Demapping>

The subcarrier demapping is performed on the complex-valued modulation symbols to obtain complex-valued modulation symbols of a corresponding physical channel. For example, the processor(s) 102 may obtain complex-valued modulation symbols mapped to subcarriers belong to PDSCH from among complex-valued modulation symbols received in a bandwidth part. For another example, the processor(s) 202 may obtain complex-valued modulation symbols mapped to subcarriers belong to PUSCH from among complex-valued modulation symbols received in a bandwidth part.

<Transform De-Precoding>

Transform de-precoding (e.g. IDFT) is performed on the complex-valued modulation symbols of the uplink physical channel if the transform precoding has been enabled for the uplink physical channel. For the downlink physical channel and for the uplink physical channel for which the transform precoding has been disabled, the transform de-precoding is not performed.

<Layer Demapping>

The complex-valued modulation symbols are de-mapped into one or two codewords.

<Demodulation and Descrambling>

The complex-valued modulation symbols of a codeword are demodulated and de-scrambled into bits of the codeword.

<Decoding>

The codeword is decoded into a transport block. For UL-SCH and DL-SCH, either LDPC base graph 1 or 2 is selected based on the size of the transport block and coding rate *R*. The codeword may include one or multiple coded blocks. Each coded block is decoded with the selected LDPC base graph into a CRC-attached code block or CRC-attached transport block. If code block segmentation was performed on a CRC-attached transport block at the transmitting side, a CRC sequence is removed from each of CRC-attached code blocks, whereby code blocks are obtained. The code blocks are concatenated into a CRC-attached transport block. The transport block CRC sequence is removed from the CRC-attached transport block, whereby the transport block is obtained. The transport block is delivered to the MAC layer.

In the above described physical layer processing at the transmitting and receiving sides, the time and frequency domain resources (e.g. OFDM symbol, subcarriers, carrier

frequency) related to subcarrier mapping, OFDM modulation and frequency up/down conversion can be determined based on the resource allocation (e.g., UL grant, DL assignment).

For uplink data transmission, the processor(s) **102** of the present disclosure may apply (or control the transceiver(s) **106** to apply) the above described physical layer processing of the transmitting side to the data unit of the present disclosure to transmit the data unit wirelessly. For downlink data reception, the processor(s) **102** of the present disclosure may apply (or control the transceiver(s) **106** to apply) the above described physical layer processing of the receiving side to received radio signals to obtain the data unit of the present disclosure.

For downlink data transmission, the processor(s) **202** of the present disclosure may apply (or control the transceiver(s) **206** to apply) the above described physical layer processing of the transmitting side to the data unit of the present disclosure to transmit the data unit wirelessly. For uplink data reception, the processor(s) **202** of the present disclosure may apply (or control the transceiver(s) **206** to apply) the above described physical layer processing of the receiving side to received radio signals to obtain the data unit of the present disclosure.

FIG. **10** illustrates operations of the wireless devices based on the implementations of the present disclosure.

The first wireless device **100** of FIG. **2** may generate first information/signals according to the functions, procedures, and/or methods described in the present disclosure, and then transmit radio signals including the first information/signals wirelessly to the second wireless device **200** of FIG. **2** (**S10**). The first information/signals may include the data unit(s) (e.g. PDU, SDU, RRC message) of the present disclosure. The first wireless device **100** may receive radio signals including second information/signals from the second wireless device **200** (**S30**), and then perform operations based on or according to the second information/signals (**S50**). The second information/signals may be transmitted by the second wireless device **200** to the first wireless device **100** in response to the first information/signals. The second information/signals may include the data unit(s) (e.g. PDU, SDU, RRC message) of the present disclosure. The first information/signals may include contents request information, and the second information/signals may include contents specific to the usage of the first wireless device **100**. Some examples of operations specific to the usages of the wireless devices **100** and **200** will be described below.

In some scenarios, the first wireless device **100** may be a hand-held device **100d** of FIG. **1**, which performs the functions, procedures, and/or methods described in the present disclosure. The hand-held device **100d** may acquire information/signals (e.g., touch, text, voice, images, or video) input by a user, and convert the acquired information/signals into the first information/signals. The hand-held devices **100d** may transmit the first information/signals to the second wireless device **200** (**S10**). The second wireless device **200** may be any one of the wireless devices **100a** to **100f** in FIG. **1** or a BS. The hand-held device **100d** may receive the second information/signals from the second wireless device **200** (**S30**), and perform operations based on the second information/signals (**S50**). For example, the hand-held device **100d** may output the contents of the second information/signals to the user (e.g. in the form of text, voice, images, video, or haptic) through the I/O unit of the hand-held device **100d**.

In some scenarios, the first wireless device **100** may be a vehicle or an autonomous driving vehicle **100b**, which

performs the functions, procedures, and/or methods described in the present disclosure. The vehicle **100b** may transmit (**S10**) and receive (**S30**) signals (e.g. data and control signals) to and from external devices such as other vehicles, BSs (e.g. gNBs and road side units), and servers, through its communication unit (e.g. communication unit **110** of FIG. **1C**). The vehicle **100b** may include a driving unit, and the driving unit may cause the vehicle **100b** to drive on a road. The driving unit of the vehicle **100b** may include an engine, a motor, a powertrain, a wheel, a brake, a steering device, etc. The vehicle **100b** may include a sensor unit for acquiring a vehicle state, ambient environment information, user information, etc. The vehicle **100b** may generate and transmit the first information/signals to the second wireless device **200** (**S10**). The first information/signals may include vehicle state information, ambient environment information, user information, and etc. The vehicle **100b** may receive the second information/signals from the second wireless device **200** (**S30**). The second information/signals may include vehicle state information, ambient environment information, user information, and etc. The vehicle **100b** may drive on a road, stop, or adjust speed, based on the second information/signals (**S50**). For example, the vehicle **100b** may receive map the second information/signals including data, traffic information data, etc. from an external server (**S30**). The vehicle **100b** may generate an autonomous driving path and a driving plan based on the second information/signals, and may move along the autonomous driving path according to the driving plan (e.g., speed/direction control) (**S50**). For another example, the control unit or processor(s) of the vehicle **100b** may generate a virtual object based on the map information, traffic information, and vehicle position information obtained through a GPS sensor of the vehicle **100b** and an I/O unit **140** of the vehicle **100b** may display the generated virtual object in a window in the vehicle **100b** (**S50**).

In some scenarios, the first wireless device **100** may be an XR device **100c** of FIG. **1**, which performs the functions, procedures, and/or methods described in the present disclosure. The XR device **100c** may transmit (**S10**) and receive (**S30**) signals (e.g., media data and control signals) to and from external devices such as other wireless devices, hand-held devices, or media servers, through its communication unit (e.g. communication unit **110** of FIG. **1C**). For example, the XR device **100c** transmits content request information to another device or media server (**S10**), and download/stream contents such as films or news from another device or the media server (**S30**), and generate, output or display an XR object (e.g. an AR/VR/MR object), based on the second information/signals received wirelessly, through an I/O unit of the XR device (**S50**).

In some scenarios, the first wireless device **100** may be a robot **100a** of FIG. **1**, which performs the functions, procedures, and/or methods described in the present disclosure. The robot **100a** may be categorized into an industrial robot, a medical robot, a household robot, a military robot, etc., according to a used purpose or field. The robot **100a** may transmit (**S10**) and receive (**S30**) signals (e.g., driving information and control signals) to and from external devices such as other wireless devices, other robots, or control servers, through its communication unit (e.g. communication unit **110** of FIG. **1C**). The second information/signals may include driving information and control signals for the robot **100a**. The control unit or processor(s) of the robot **100a** may control the movement of the robot **100a** based on the second information/signals.

In some scenarios, the first wireless device **100** may be an AI device **400** of FIG. 1. The AI device may be implemented by a fixed device or a mobile device, such as a TV, a projector, a smartphone, a PC, a notebook, a digital broadcast terminal, a tablet PC, a wearable device, a Set Top Box (STB), a radio, a washing machine, a refrigerator, a digital signage, a robot, a vehicle, etc. The AI device **400** may transmit (S10) and receive (S30) wired/radio signals (e.g., sensor information, user input, learning models, or control signals) to and from external devices such as other AI devices (e.g., **100a**, . . . , **100f**, **200**, or **400** of FIG. 1) or an AI server (e.g., **400** of FIG. 1) using wired/wireless communication technology. The control unit or processor(s) of the AI device **400** may determine at least one feasible operation of the AI device **400**, based on information which is determined or generated using a data analysis algorithm or a machine learning algorithm. The AI device **400** may request that external devices such as other AI devices or AI server provide the AI device **400** with sensor information, user input, learning models, control signals and etc. (S10). The AI device **400** may receive second information/signals (e.g., sensor information, user input, learning models, or control signals) (S30), and the AI device **400** may perform a predicted operation or an operation determined to be preferred among at least one feasible operation based on the second information/signals (S50).

Hereinafter, Buffer Status reporting (BSR) procedure in the NR system is described.

The BSR procedure is used to provide the serving gNB with information about UL data volume in the MAC entity. RRC configures the following parameters to control the BSR:

periodicBSR-Timer;
retxBSR-Timer;
logicalChannelSR-DelayTimerApplied;
logicalChannelSR-DelayTimer;
logicalChannelSR-Mask;
logicalChannelGroup.

Each logical channel may be allocated to an LCG using the logicalChannelGroup. The maximum number of LCGs is eight. The MAC entity determines the amount of UL data available for a logical channel according to the data volume calculation procedure.

Hereinafter, Logical channel prioritization in the NR system is described.

The Logical Channel Prioritization procedure is applied when a new transmission is performed. RRC controls the scheduling of uplink data by signalling for each logical channel: priority where an increasing priority value indicates a lower priority level, prioritisedBitRate which sets the Prioritized Bit Rate (PBR), bucketSizeDuration which sets the Bucket Size Duration (BSD).

The MAC entity shall maintain a variable B_j for each logical channel j . B_j shall be initialized to zero when the related logical channel is established, and incremented by the product $PBR \times TTI$ duration for each TTI, where PBR is Prioritized Bit Rate of logical channel j . However, the value of B_j can never exceed the bucket size and if the value of B_j is larger than the bucket size of logical channel j , it shall be set to the bucket size. The bucket size of a logical channel is equal to $PBR \times BSD$, where PBR and BSD are configured by upper layers.

The MAC entity shall perform the following Logical Channel Prioritization procedure when a new transmission is performed:

The MAC entity shall allocate resources to the logical channels in the following steps 1-3:

Step 1: All the logical channels with $B_j > 0$ are allocated resources in a decreasing priority order. If the PBR of a logical channel is set to "infinity", the MAC entity shall allocate resources for all the data that is available for transmission on the logical channel before meeting the PBR of the lower priority logical channel(s);

Step 2: the MAC entity shall decrement B_j by the total size of MAC SDUs served to logical channel j in Step 1. Especially, the value of B_j can be negative.

Step 3: if any resources remain, all the logical channels are served in a strict decreasing priority order (regardless of the value of B_j) until either the data for that logical channel or the UL grant is exhausted, whichever comes first. Logical channels configured with equal priority should be served equally.

The UE shall also follow the rules (1)-(4) below during the scheduling procedures above

(1) the UE should not segment an RLC SDU (or partially transmitted SDU or re-transmitted RLC PDU) if the whole SDU (or partially transmitted SDU or re-transmitted RLC PDU) fits into the remaining resources of the associated MAC entity.

(2) if the UE segments an RLC SDU from the logical channel, it shall maximize the size of the segment to fill the grant of the associated MAC entity as much as possible.

(3) the UE should maximise the transmission of data.

(4) if the MAC entity is given an UL grant size that is equal to or larger than 4 bytes while having data available for transmission, the MAC entity shall not transmit only padding BSR and/or padding.

The MAC entity shall not generate a MAC PDU for the HARQ entity if the following conditions are satisfied:

the MAC entity is configured with skipUplinkTxDynamic with value true and the grant indicated to the HARQ entity was addressed to a C-RNTI, or the grant indicated to the HARQ entity is a configured uplink grant; and

there is no aperiodic CSI requested for this PUSCH transmission; and

the MAC PDU includes zero MAC SDUs; and
the MAC PDU includes only the periodic BSR and there is no data available for any LCG, or the MAC PDU includes only the padding BSR.

Logical channels shall be prioritised in accordance with the following order (highest priority listed first):

- C-RNTI MAC CE or data from UL-CCCH;
- Configured Grant Confirmation MAC CE;
- MAC CE for BSR, with exception of BSR included for padding;
- Single Entry PHR MAC CE or Multiple Entry PHR MAC CE;
- data from any Logical Channel, except data from UL-CCCH;
- MAC CE for Recommended bit rate query;
- MAC CE for BSR included for padding.

In the LTE system and the NR system, Logical channel group (LCG) is used to trigger and report a buffer size regarding a logical channel. In specific, if a logical channel belongs to an LCG, the new data arrival for that logical channel triggers a BSR whereas a logical channel not belonging to any LCG does not trigger a BSR even if the new data arrives for that logical channel. The reason was that:

The logical channel not belonging to any LCG is likely to be of lower priority so that there is no need to trigger a BSR. For these kind of lower priority logical chan-

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nels, it is sufficient to transmit the data when the received UL grant remains after including all higher priority logical channel data.

The scheduler may be able to know the characteristics of the traffic, e.g., data size, periodicity, so that the UE waits until it is scheduled rather than triggering or reporting a BSR unnecessarily.

While, IAB (Integrated access and backhaul) based radio access network (RAN) architecture consists of one or more IAB nodes, which support wireless access to UEs and wirelessly backhauls the access traffic, and one or more IAB donors which provide UE's interface to core network and wireless backhauling functionality to IAB nodes.

FIG. 11 shows an example of IAB based RAN architectures.

Especially, in FIG. 11, the IAB based RAN architectures consist of one or more IAB nodes, which support wireless access to UEs and wirelessly backhauls the access traffic, and one or more IAB donors which provide UE's interface to core network and wireless backhauling functionality to IAB nodes. Each adaptation layer of these IAB-nodes and IAB donors carries the following information in order to identify UE and/or radio bearer for control-plane or user-plane data:

- UE-bearer-specific ID
- UE-specific ID
- Route ID, IAB-node or IAB-donor address
- QoS information

In NR MAC layer, there are two types of BSR format. One is for Short BSR and another is for Long BSR. If more than one LCG has data available for transmission when a MAC PDU containing a BSR is to be built, a MAC entity shall report Long BSR for all LCGs which have data available for transmission, but if only one LCG has data available for transmission, Short BSR shall be reported.

In this condition, if Long BSR is reported, 1 byte LCG field should be always included into the Long BSR format regardless of whether how many LCGs have data available for transmission, i.e., LCG field length is fixed to be able to contain maximum LCG ID.

In NRIAB system, unified design was selected to support 1:1 and N:1 bearer mapping together. However, to support 1:1 bearer mapping in IAB, each logical channel (LoCH) should be associated with one bearer at every hop over the path. This means that if an IAB node supports 100 UEs and each UE has 25 bearers, the IAB node should have at least 250 LoCHs to support 1:1 bearer mapping. With this understanding, RAN2 identified that the current LCID space and LCG space would not be enough to support 1:1 bearer mapping in IAB as shown in 3GPP TR 38.874.

Thus, if LCG space increase, a new BSR format should be defined. If a new BSR format follows current BSR format and possible LCG ID increases up to 23.3 byte LCG field should be always included into the BSR format as there may be 24 configured LCGs.

Unfortunately, however, it is hard to determine whether how many LCID and LCG spaces is needed and the required size of LCID space and LCG space can be determined by operator's design. For instance, if small size of an IAB network is considered, 7 is enough for maximum LCG ID, but if operator wants large size of an IAB network, 24 or larger LCG ID and even much larger LCID space would be required. This means that anyway LCG space should be defined as large as possible to support big enough size of IAB network which operators want and a maximum LCG ID can be differently configured according to network design and placement of IAB node.

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Considering this, a new BSR format to support the extended LCG ID should be introduced and it should be also defined when the new BSR format for the extended LCG ID is used because if a new BSR format for the extended LCG ID is always used, unnecessary overhead cannot be avoided, e.g., if maximum LCG ID is 23, the MAC entity has to always use 3 byte LCG field even though only 8 LCGs are configured and used.

Thus, this overhead should be avoided and a new BSR format and method need to be considered.

According to the present disclosure, for supporting LCG ID extension, it is suggested that an extended BSR (eBSR) should consist of variable size of a LCG field and corresponding Buffer Size field. A UE or IAB node (or a MAC entity at the UE/IAB node) determines the size of the LCG field based on the maximum LCG ID value among the LCG IDs of the LCG which are configured to the UE (or the MAC entity at the UE) by a network.

In the following description, it is assumed that an integer among 0 to the maximum LCG-ID is assigned as an LCG ID for an LCG by a network.

The extended BSR format consists of Buffer Size field, where Buffer Size field may be one byte per corresponding LCG. The extended BSR format consists of variable size of a LCG field.

The size of a LCG field can be from 1 byte to X bytes, where the X bytes depends on the maximum value of LCG ID, which can be configured to the MAC entity by the network (through layer-2 (L2) or layer-3 (L3) signaling). For example, if the maximum value of LCG ID is 31, X is equal to 4. Further, if the maximum value of LCG ID is 15, X is equal to 2.

The MAC entity determines the actual size of the LCG field to be included in the eBSR format based on the maximum value of the LCG ID among the LCG IDs of the LCG which are actually configured to the MAC entity. That is, a number of the LCG IDs is variable according to a number of node served by the MAC entity. Or, the number of the LCG IDs is variable according to number flow or a number of radio bearer served by the MAC entity. The maximum value of LCG ID, which can be configured to the MAC entity, can be different from the maximum value of the LCG ID among the LCG IDs of the LCGs which are actually configured to the MAC entity by the network. In other words, the maximum number of LCGs, which can be configured to the MAC entity, can be different from the maximum number of LCGs which are actually configured to the MAC CE. The maximum value of LCG ID, which can be configured to the MAC entity may be predefined in the system (e.g. as system standards) or configured by the network.

For example, the maximum value of the LCG ID among the LCG IDs of the LCGs, which are actually configured to the MAC entity, can be 7 where as the maximum value of LCG ID, which can be configured to the MAC entity, is 31.

Further, the size of LCG field is defined as $\lceil \text{floor}(Y/8) \text{ plus } 1 \rceil$ bytes, if the maximum value of the LCG ID among the LCG IDs of the LCGs, which are actually configured to the MAC entity, is equal to or larger than Y and less than Y+8, where Y is multiple of 8, i.e., 0, 8, 16, 24, etc.

For example, the size of LCG field is one byte if the maximum value of the LCG ID among the LCG IDs of the LCGs, which are actually configured to the MAC entity, is equal to or larger than 0 and less than 8. Or, the size of LCG field is two bytes if the maximum value of the LCG ID

among the LCG IDs of the LCGs, which are actually configured to the MAC entity, is equal to or larger than 8 and less than 16.

Similarly, the size of LCG field is three bytes if the maximum value of the LCG ID among the LCG IDs of the LCGs, which are actually configured to the MAC entity, is equal to or larger than 16 and less than 24. The size of LCG field is four bytes if the maximum value of the LCG ID among the LCG IDs of the LCGs, which are actually configured to the MAC entity, is equal to or larger than 24 and less than 32;

Meanwhile, the eBSR format can be used for Long BSR or Long Truncated BSR.

An LCG field contains a bitmap and each bit in the bitmap is associated with LCG *i*.

For Long BSR, the MAC entity indicates the presence of the Buffer Size field corresponding to logical channel group *i* by LCG_{*i*} of the LCG field. For Long Truncated BSR, the MAC entity indicates whether the logical channel group *i* has data available by LCG_{*i*} of the LCG field.

FIG. 12 shows examples of eBSR format according to the present disclosure.

Firstly, referring to FIG. 12(a), if the maximum value of the LCG ID among the LCG IDs of the LCGs, which are actually configured to the MAC entity, is equal to or larger than 0 and less than 8, the LCG field is one byte.

Next, referring to FIG. 12(b), if the maximum value of the LCG ID among the LCG IDs of the LCGs, which are actually configured to the MAC entity, is equal to or larger than 8 and less than 16, the LCG field is two bytes.

Finally, referring to FIG. 12(c), if the maximum value of the LCG ID among the LCG IDs of the LCGs, which are actually configured to the MAC entity, is equal to or larger than 16 and less than 24, the LCG field is three bytes.

In the above description, the MAC entity may be a MAC entity at a UE, a MAC entity at an IAB node, or a MAC entity at a network node (e.g., BS).

FIG. 13 shows an example of a BSR procedure according to the present disclosure.

Referring to FIG. 13, a node (e.g., BS or IAB node) at a network side may transmit information on LCG ID configuration to a UE side (e.g. UE or another IAB node) at S101.

The UE/IAB node at the UE side may determine the length for a LCG field to be included in a BSR MAC CE at S102, based on a maximum LCG ID value which is (actually) configured to the UE/IAB node at the UE side.

The UE/IAB node at the UE side may generate a BSR MAC CE if there is a BSR to be transmitted at S103. In the invention, the UE/IAB node at the UE side generates the BSR MAC CE to include a LCG field having the length determined at S102.

The UE/IAB node at the UE side transmits the BSR MAC CE. The BSR MAC CE is included in a MAC PD U to be transmitted over a physical uplink channel (e.g. physical uplink shared channel (PUSCH) at S104.

When the above suggestion of the present disclosure is applied, the UE/IAB node (more specifically, MAC entity at the UE/IAB node) behavior is as follows.

A UE or IAB node is configured with at least one LCG, which is associated with an LCG ID by receiving a configuration of LCG via L2 or L3 signaling. The L2 or L3 signaling can be one of the MAC, RLC, PDCP, or RRC signaling.

The MAC entity determines the size of LCG field by checking the maximum value of the LCG ID among the LCG IDs of the LCG which are actually configured to the MAC entity.

In BSR operation, a BSR trigger condition may be the same as legacy BSR conditions. The MAC entity triggers a BSR if any of the following events occur:

- 1) the MAC entity has new UL data available for a logical channel which belongs to an LCG. the new UL data belongs to a logical channel with higher priority than the priority of any logical channel containing available UL data which belong to any LCG, or none of the logical channels which belong to an LCG contains any available UL data. In this case, the triggered BSR is referred to as Regular BSR.
- 2) UL resources are allocated and number of padding bits is equal to or larger than the size of the Buffer Status Report MAC CE plus its subheader, in which case the BSR is referred to as Padding BSR;
- 3) retxBSR-Timer expires, and at least one of the logical channels which belong to an LCG contains UL data, in which case the BSR is referred below to as Regular BSR;
- 4) periodicBSR-Timer expires, in which case the BSR is referred to as Periodic BSR.

When the BSR is triggered, the MAC entity according to an example of the present disclosure shall report Long BSR or eBSR for all LCGs which have data available for transmission, if more than one LCG has data available for transmission when the MAC PDU containing the BSR is to be built:

While a BSR is pending, if the MAC entity generates a BSR MAC CE using the eBSR format, the MAC entity includes LCG field having the size, which is determined as above, i.e., the maximum value of the LCG ID among the LCG IDs of the LCG which are actually configured to the MAC entity.

After generating the BSR MAC CE, the MAC entity transmits the generated BSR MAC CE. Then the MAC entity may receive an UL grant to transmit the UL data available for a logical channel in response to the transmitted BSR MAC CE, and transmits the UL data from the logical channel which has data available for transmission to the network using the received UL grant.

If the maximum value of the LCG ID among the LCG IDs of the LCGs, which are actually configured to the MAC entity, is equal to or larger than 0 and less than 8, the MAC entity can choose one of legacy Long BSR format having one byte LCG field or eBSR format having one byte LCG field.

Alternatively L field which indicates the size of LCG field can be included into the BSR MAC CE, i.e., if the value of L field is 1, one byte of LCG field is used and if the value of L field is 2, two bytes of LCG field is used. The value of the L field and the size of LCG field are determined by the maximum value of the LCG ID among the LCG IDs of the LCG which are actually configured to the MAC entity.

When this invention is applied, the network behavior is as follows:

A network node (e.g. BS) may transmit the L2 or L3 signaling including the configuration of LCG for configuring a MAC entity for a UE or IAB node. In the following description, an MAC entity for a UE or IAB node is referred to as a UE MAC entity. A UE MAC entity is configured in a UE, and may be configured in an IAB node and/or in a network node (e.g. BS).

When a network (e.g. BS) receives a BSR MAC CE using the eBSR format after configuring a UE MAC entity with at least one LCG, the network can determine the size of LCG field in the BSR MAC CE using the eBSR format based on

the maximum value of the LCG ID among the LCG IDs of the LCG which the network actually configures to the UE MAC entity.

For example, when the maximum value of LCG ID, which the network can configure to the UE MAC entity, is 31. If the maximum value of the LCG ID among the LCG IDs of the LCGs, which the network actually configures to the UE MAC entity, is 15, the size of LCG field in the BSR MAC CE using the eBSR format is two bytes.

If the maximum value of the LCG ID among the LCG IDs of the LCGs, which the network actually configures to the UE MAC entity, is 23, the size of LCG field in the BSR MAC CE using the eBSR format is three bytes.

If the maximum value of the LCG ID among the LCG IDs of the LCGs, which the network actually configures to the UE MAC entity, is 31, the size of LCG field in the BSR MAC CE using the eBSR format is four bytes.

In response to the received BSR MAC CE using the eBSR, the network provides an UL grant to the UE, and receives the MAC PDU including user data via the provided UL grant.

FIG. 14 shows another example of a BSR procedure according to the present disclosure. In FIG. 14, it is assumed that a network can configure LCG ID up to 31, but actually configures LCG ID of a MAC entity with 15.

Even though the network can configure LCG ID=31, a UE/IAB node (e.g. a MAC entity at the UE/IAB node) is configured with LCG ID=15 by the network (S201). The MAC entity determines the two byte long of LCG field for a BSR MAC CE (S202) because the maximum value of the LCG ID among the LCG IDs of the LCG which are actually configured to the MAC entity is 15.

The MAC entity triggers a BSR (S203) and while generating the BSR MAC CE after receiving a UL grant (204), the MAC entity includes two bytes of LCG field into the BSR MAC CE (S205) because the maximum value of the LCG ID among the LCG IDs of the LCG which are actually configured to the MAC entity is 15.

After generating the BSR MAC CE including two byte of LCG field and the associated Buffer Size information, the MAC entity transmits the generated BSR MAC CE using the received UL grant (S206).

According to the present disclosure, when LCG space is extended, the required additional LCG space can be different according to the placement of IAB node. For example, if an IAB node is connected to the IAB donor and has lots of child IAB nodes, this IAB node should have large number of LCG space to support lots of UE bearers, but if an IAB node is placed at the edge of the IAB network and needs to support only a few UEs and UE bearers, current LCG space, i.e., 7, would be enough. In this condition, even though each IAB node can be configured with different maximum LCG ID, if all IAB node in an IAB networks use same extended BSR format for the extended LCG ID, most of all IAB nodes only except for the IAB nodes where are close to the IAB donor would have unnecessary overhead whenever reporting BSR.

Further, according to the present disclosure, each IAB node can determine LCG field length based on the configured maximum LCG ID. For example, if 7 is the configured maximum LCG ID, 1 byte LCG field length would be selected for maximum 8 LCGs, which brings the same amount of LCG field overhead as legacy BSR MAC CE format. But if 23 is the configured maximum LCG ID, 3 byte LCG field length would be selected for maximum 24 LCGs. Thereby, each IAB node would be able to avoid unnecessary

overhead and select appropriate LCG field length when a BSR is reported, even though the configured maximum LCGID is different.

According to the implementations of the present disclosure, a UE performs measurements after transmitting or receiving URLLC traffics during a configured measurement gap, thereby satisfying the requirement for the URLLC service while guaranteeing the UE coverage.

What is claimed is:

1. A method comprising:

receiving, by an Integrated Access and Backhaul (IAB) node from a network, configuration information related to logical channel groups (LCGs);

determining, by the IAB node, a length of a LCG field based on a maximum LCG identifier configurable for IAB node;

generating, by the IAB node, a buffer status report (BSR) including the LCG field having the determined length; and

transmitting, by the IAB node to the network, the BSR, wherein the length of the LCG field is determined to floor(Y/8) plus 1 bytes,

wherein the maximum LCG identifier is equal to or larger than Y and less than Y+8, and

wherein Y is multiple of 8.

2. The method of claim 1, wherein each bit of the LCG field indicates whether buffer size information of a corresponding LCG is present or not in the BSR.

3. The method of claim 1, wherein the maximum LCG identifier is variable according to a number of nodes served by the IAB node.

4. The method of claim 1, wherein the maximum LCG identifier is variable according to a number flows or a number of radio bearers served by the IAB node.

5. An Integrated Access and Backhaul (IAB) node comprising:

at least one processor; and

at least one computer memory storing instructions which, when executed by the at least one processor, cause the IAB node to perform operations comprising:

receiving, from a network, configuration information related to logical channel groups (LCGs);

determining a length of a LCG field based on a maximum LCG identifier configurable for IAB node;

generating a buffer status report (BSR) including the LCG field having the determined length; and

transmitting the BSR to the network, wherein the length of the LCG field is determined to floor(Y/8) plus 1 bytes,

wherein the maximum LCG identifier is equal to or larger than Y and less than Y+8, and

wherein Y is multiple of 8.

6. The IAB node of claim 5, wherein each bit of the LCG field indicates whether buffer size information of a corresponding LCG is present or not in the BSR.

7. The IAB node of claim 5, wherein the maximum LCG identifier is variable according to a number of nodes served by the IAB node.

8. The IAB node of claim 5, wherein the maximum LCG identifier is variable according to a number flows or a number of radio bearers served by the IAB node.

9. An apparatus for an Integrated Access and Backhaul (IAB) node, the apparatus comprising:

at least one processor; and

at least one computer memory storing instructions that, when executed by the at least one processor, cause the IAB node to perform operations comprising:

receiving, from a network, configuration information
 related to logical channel groups (LCGs);
 determining a length of a LCG field based on maximum
 LCG identifier configurable for IAB node;
 generating a buffer status report (BSR) including the LCG 5
 field having the determined length; and
 transmitting the BSR to the network,
 wherein the length of the LCG field is determined to
 floor($Y/8$) plus 1 bytes,
 wherein the maximum LCG identifier is equal to or larger 10
 than Y and less than $Y+8$, and
 wherein Y is multiple of 8.

10. A non-transitory computer readable storage medium
 comprising program instructions that, when executed by at
 least one processor, cause an Integrated Access and Back- 15
 haul (IAB) node to perform operations comprising:

receiving, from a network, configuration information
 related to logical channel groups (LCGs);
 determining a length of a LCG field based on a maximum 20
 LCG identifier for IAB node;
 generating a buffer status report (BSR) including the LCG
 field having the determined length; and
 transmitting the BSR to a network,
 wherein the length of the LCG field is determined to
 floor($Y/8$) plus 1 bytes, 25
 wherein the maximum LCG identifier is equal to or larger
 than Y and less than $Y+8$, and
 wherein Y is multiple of 8.

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